Does Machine Learning Improve on Simple Tools for Osteoporosis Screening? Temporal Validation, Calibration and Subgroup Performance in NHANES

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Abstract

Background. Machine learning (ML) models for osteoporosis screening are often built on survey data but are rarely compared with simple clinical tools under equal conditions.

Methods. We studied 9,657 adults aged ≥50 years from five NHANES cycles not diagnosed with or treated for osteoporosis; the outcome was a hip T-score ≤ −2.5. Models were developed on 2005–2014 (343 cases) and validated on 2017–2018 data (n = 2,008; 96 cases). Eight model families using 35 predictors from history, examination and blood tests were each tuned with 40 Bayesian-optimisation trials and recalibrated. In a testing hierarchy, the primary comparison was the best model versus the Osteoporosis Self-Assessment Tool (OST); secondary comparisons, Holm-adjusted, were with ORAI, SCORE, an age–weight–sex model and a Boruta + XGBoost pipeline. ML families were tested for equivalence with logistic regression (margin ±0.02 in AUROC).

Results. The selected penalised logistic regression had a validation AUROC of 0.869 (95% CI 0.831–0.904), with a calibration slope of 1.00. It exceeded OST (0.841) by 0.028 (0.007–0.052; $P = 0.014$), but the interval included zero when refitting was bootstrapped (−0.004 to 0.052). Across five held-out cycles, the pooled gain over OST was 0.025 (0.015–0.034). The model exceeded ORAI (0.791) but not SCORE (0.841; adjusted $P = 0.11$) or the age–weight–sex model (0.851; adjusted $P = 0.21$). Six of seven ML families were equivalent to logistic regression. A 10-predictor short form without blood tests performed like the age–weight–sex model (0.852).

Conclusions. Equally tuned ML did not outperform logistic regression, and 35 predictors added about 0.02–0.03 in AUROC to simple tools, a gain of uncertain clinical value.

1 Introduction

Osteoporosis is a systemic skeletal disease of low bone mass and deteriorated bone microarchitecture that leads to fragility fractures [1]. Among US adults aged 50 years and older, about one in eight has osteoporosis at the femoral neck or lumbar spine, and its prevalence in women is several times that in men [2]. Hip and vertebral fractures cause disability, loss of independence and excess mortality. Yet effective treatments can lower fracture risk substantially once osteoporosis is recognised [1].

Diagnosis depends on dual-energy X-ray absorptiometry (DXA), with osteoporosis defined as a T-score of −2.5 or lower [3]. DXA is not universally available. Guidelines target screening mainly at women aged 65 and older, leaving men and younger high-risk adults under-assessed [4]. Simple tools that use routine information to prioritise who most needs DXA could help close this gap. The Osteoporosis Self-Assessment Tool (OST), $0.2 \times (\text{weight} - \text{age})$, and its Asian counterpart OSTA are widely used for this purpose [5, 6]. Other questionnaire-based indices include the Osteoporosis Risk Assessment Instrument (ORAI) and the Simple Calculated Osteoporosis Risk Estimation (SCORE) [7, 8]. FRAX instead estimates fracture probability from clinical risk factors [9].

Machine learning (ML) can combine many routine predictors, and several studies have applied it to survey data such as the National Health and Nutrition Examination Survey (NHANES). Most recently, Zhang et al. compared three feature-selection methods (Boruta, XGBoost and Lasso) with four classifiers on NHANES, reporting performance mainly as accuracy, with the highest area under the receiver operating characteristic curve (AUROC, 0.85) for

random forest [10]. Such studies, however, leave several questions unanswered:

1. **Is ML better than existing simple tools?** ML models are rarely benchmarked against OST, ORAI, SCORE or a minimal regression model, and the ML models are often tuned more carefully than the regression they are compared with, or the reverse.
2. **Is the evaluation informative?** When most people do not have osteoporosis, high accuracy can be achieved by predicting "no osteoporosis" for everyone. Calibration and clinical net benefit matter more for screening [11, 12].
3. **Is the target population right?** A case-finding tool is meant for people not yet diagnosed or treated, yet survey-based models usually include participants who already know they have osteoporosis and may be treated for it.
4. **Does the model hold up over time?** Random splits from the same years do not show whether performance persists in later populations.
5. **Is the model equitable?** Pooled models can underperform in men or specific racial and ethnic groups, and using race as a predictor is increasingly questioned [13, 14].

Evidence from other prediction tasks suggests that ML often does not outperform logistic regression [15], and that class-imbalance corrections can harm calibration [16]. Our aim was not to propose another model, but to ask how much ML adds to simple tools for osteoporosis case-finding when every method is tuned equally and evaluated in the intended population and over time.

We addressed these questions using 9,657 adults aged 50 and older from five NHANES cycles who had not been diagnosed with or treated for osteoporosis. Our contributions are:

- a temporal validation of eight equally tuned model families against three established screening indices (OST, ORAI and SCORE), a three-variable logistic model and a Boruta + XGBoost pipeline, in a screening population, with an explicit hierarchy of hypotheses and an equivalence test for ML versus regression;
- internal–external cross-validation across all five survey cycles, and a bootstrap that includes uncertainty from model refitting;
- a comparison of three class-imbalance strategies, with recalibration;
- a quantification of how much each group of predictors adds beyond age, weight and sex;
- an evaluation of discrimination, calibration and net benefit with confidence intervals, survey weighting and adjustment for DXA non-response;
- a subgroup analysis reporting event counts and uncertainty, including the strata targeted by screening guidelines (women aged 65 or older, younger postmenopausal women and older men), non-Hispanic Asian adults, sex-specific thresholds and a race-free model;
- a short form using only clinically obtainable, race-free predictors, published as an explicit equation, with a discussion of implementation in electronic health records.

The study is reported according to TRIPOD+AI [17] (Supplementary Table S17).

2 Methods

2.1 Data source

This study used publicly available, de-identified data from NHANES, a cross-sectional survey of the non-institutionalised US civilian population with a complex, multistage probability design, conducted by the National Center for Health Statistics (NCHS) [18]. NHANES combines household interviews with standardised physical examinations, DXA and laboratory testing in mobile examination centres. We pooled the five 2-year cycles with femoral DXA measurements: 2005–2006, 2007–2008, 2009–2010, 2013–2014 and 2017–2018. The following files were downloaded from the NCHS website and merged on the respondent sequence number:

- demographic and body-measure files;
- DXA files for the femur and lumbar spine;
- laboratory files for standard biochemistry and vitamin D;
- questionnaire files for smoking, diabetes, medical conditions, osteoporosis history, reproductive health and alcohol use;
- prescription-medication files (used only to identify osteoporosis treatment).

2.2 Study population

Of 50,463 participants in the five cycles, 14,166 were aged 50 years or older, and 10,981 of these had a valid femoral neck bone mineral density (BMD) measurement. Participants missing more than 50% of candidate predictors were to be excluded, but none met this criterion. Because the intended use is case-finding before DXA, we then excluded 1,324 participants who reported having been told by a doctor that they had osteoporosis (1,155) or who reported current use of an osteoporosis medication without such a diagnosis (169). Medications were bisphosphonates (alendronate, risedronate, ibandronate, zoledronic acid, etidronate), denosumab, raloxifene, bazedoxifene, teriparatide, abaloparatide, romosozumab and calcitonin; 486 excluded participants used one. Of the 1,324 excluded participants, 282 met the outcome definition. The final cohort comprised 9,657 participants (Figure 1). Participants with a missing answer to the diagnosis question were treated as not diagnosed. The 3,185 participants without DXA are compared with the cohort in Supplementary Table S8, and the analysis including diagnosed or treated participants is reported as a sensitivity analysis.

2.3 Outcome definition

The primary outcome was osteoporosis at the hip, defined according to World Health Organization criteria [3] as a T-score of -2.5 or lower at the femoral neck or total hip. T-scores were calculated as (participant BMD $-$ reference mean)/reference SD. The reference values were those of NHANES III non-Hispanic white women aged 20–29 years [19]: mean $0.858\,\mathrm{g/cm^2}$ (SD 0.120) for the femoral neck and mean $0.942\,\mathrm{g/cm^2}$ (SD 0.122) for the total hip. The same female references were applied to both sexes, as recommended by the International Society for Clinical Densitometry (ISCD) [20]. DXA was performed and analysed by trained technologists following the standard NHANES protocol, independently of the questionnaire and laboratory data used as predictors.

The primary outcome was revised during the analysis. Earlier versions of this study defined osteoporosis at the femoral neck or lumbar spine. We changed to a hip-based outcome because the hip was scanned and analysed with the same protocol and reference values in every cycle, whereas the lumbar spine was not. NHANES reports a total L1–L4 BMD only for complete spine scans, and the share of partial scans (with at least one invalid vertebra) rose from 15.3% in 2005–2006 to 38.1% in 2017–2018, so a spine-based outcome would have shifted in definition between the development and validation periods. The results of every version of the analysis are reported in Supplementary Table S16.

Hip DXA was performed on Hologic QDR-4500A densitometers up to 2009–2010 and on Hologic Discovery densitometers thereafter. We found no hip BMD cross-calibration between these scanner models in the NHANES DXA documentation, and applied none. Mean femoral neck BMD in women was 0.744–0.753 g/cm^2 in 2005–2010 and 0.729–0.731 g/cm^2 in 2013–2018, a difference of about 0.15 reference SD that coincides with both the scanner change and a change in the population sampled (non-Hispanic Asian adults were oversampled from 2011). Hip-based prevalence nevertheless remained stable (4.0%–4.8% by cycle), and we refitted the model on 2013–2014 alone, which used the same scanner type as the validation cycle.

Two spine-inclusive outcomes were analysed as sensitivity analyses: (i) hip or complete-scan L1–L4 spine T-score ≤ -2.5; and (ii) hip or spine T-score ≤ -2.5, with spine BMD calculated from at least two valid vertebrae as total bone mineral content divided by total projected area [21]. Spine T-scores used the Hologic young white-female reference used in previous NHANES analyses (mean 1.047 g/cm^2, SD 0.110) [2, 22].

2.4 Candidate predictors

We selected 35 candidate predictors obtainable from history, physical examination or standard blood tests. Not all of them are routinely recorded in practice (for example, income and vitamin D); this is addressed by the short form and in the Discussion. All DXA-derived variables were excluded, as were self-reported diagnosis or treatment of osteoporosis, to prevent outcome leakage.

- **Demographic:** age; sex; race/ethnicity (Mexican American, other Hispanic, non-Hispanic white [reference], non-Hispanic Black, other/multiracial); education level; family income-to-poverty ratio (PIR).
- **Anthropometric:** body mass index (BMI), weight, height and waist circumference.
- **Laboratory:** serum calcium, phosphorus, alkaline phosphatase, creatinine, albumin, uric acid, glucose, total protein, total cholesterol and 25-hydroxyvitamin D.
- **FRAX-type clinical risk factors:** prior hip, wrist or spine fracture (any cause); parental hip fracture; ever daily use of prednisone or cortisone; current smoking; alcohol use of three or more drinks per day.
- **Other history:** former smoking; postmenopausal status (women reporting no regular menstrual periods in the past 12 months, which also captures hysterectomy); ever use of female hormones; and self-reported physician-diagnosed diabetes, arthritis, coronary heart disease, stroke and cancer.

Race/ethnicity was coded from the five-category variable (RIDRETH1) available in every cycle.

Non-Hispanic Asian adults can be identified separately only from 2011 (RIDRETH3), so in the models they fall within "other/multiracial". This category therefore changed composition over time: it made up 2.9%–4.6% of participants in 2005–2010 and 17.8% in 2017–2018, when 77% of its members were non-Hispanic Asian. We evaluated non-Hispanic Asian adults as a separate subgroup and refitted the model without them.

2.5 Data preparation

Several variables needed harmonisation across cycles:

- **Age** is top-coded at 85 years in 2005–2006 and at 80 years thereafter, so it was capped at 80 years in all cycles.
- **Creatinine** from 2005–2006 was recalibrated to the standardised method as recommended by NCHS ($-0.016 + 0.978\times$ measured value).
- **Vitamin D** was taken from the liquid chromatography–tandem mass spectrometry (LC-MS/MS) or LC-MS/MS-equivalent values released for each cycle, which are named differently in 2005–2006 (LBDVIDMS) and later cycles (LBXVIDMS).
- **Alcohol:** average daily intake was calculated as drinks per drinking day multiplied by drinking days per year, divided by 365. Drinking frequency was recorded as a count per week, month or year up to 2013–2014, and as a frequency category in 2017–2018, which we converted to approximate days per year. Non-drinkers were coded 0. Because this change coincides with the validation cycle, we also refitted the model without alcohol.
- **Zero values:** values stored in the SAS transport files as numerical underflow (absolute value $< 10^{-30}$) were restored to exactly 0.

Other laboratory measurements were used as released. Their means differed between periods (for example, alkaline phosphatase 72.6 vs 82.6 U/L and albumin 4.20 vs 4.04 g/dL; Supplementary Table S7), which probably reflects changes in laboratory methods; we therefore also refitted the model without laboratory predictors. Reproductive variables were coded 0 for men. Responses of "refused" or "don't know" were set to missing, and borderline diabetes was coded as no diabetes. To detect unharmonised variables, the analysis stopped automatically if any predictor was entirely missing within a cycle. Missingness by cycle is shown in Supplementary Table S9.

2.6 Development and temporal validation sets

Participants from 2005–2014 (n = 7,649; 343 cases) formed the development set, and participants from 2017–2018 (n = 2,008; 96 cases) the temporal validation set. All model fitting, tuning, threshold selection, recalibration and short-form predictor selection used the development set only. However, the validation set was evaluated in earlier versions of this analysis, before the outcome definition and the cohort were revised (Supplementary Table S16), so it is not untouched in a strict sense.

2.7 Sample size

No formal sample-size calculation was performed; the available NHANES data were used. The development set provided 343 events, or 9.8 per candidate predictor, so all regression models

were fitted with L2 penalisation, and calibration slopes were checked in the validation set. The validation set contained 96 events, which limits the precision of differences between models; we therefore interpret confidence intervals rather than significance alone.

2.8 Missing data

Missing predictor values were imputed by iterative chained-equation imputation (scikit-learn IterativeImputer, 10 iterations) [23]. The proportion missing was 9.5% for PIR, 5.9% for vitamin D, 4.2% for alcohol and ≤4.4% for every other predictor. The outcome was complete by design. The imputer was fitted within each cross-validation training fold, so no information leaked from validation data. All predictors, including binary ones, were then standardised.

2.9 Models, class-imbalance strategies and tuning

We compared eight model families:

- logistic regression (LR) with linear terms;
- LR with cubic B-spline terms for the 17 continuous predictors (4 knots at quantiles, linear extrapolation beyond the boundary knots; LR-spline) [24];
- a support vector machine (SVM, radial basis kernel);
- random forest (RF);
- XGBoost [25], LightGBM [26] and CatBoost [27];
- a stacking ensemble, with RF, XGBoost, LightGBM and CatBoost as base learners and a logistic regression meta-learner.

Each model was first fitted with default hyperparameters and three class-imbalance strategies: none, cost-sensitive class weighting and SMOTE [28], with resampling applied to training folds only. (SMOTE-Tomek [29], used in an earlier version, removed no Tomek links in these data and was therefore identical to SMOTE.) All 35 predictors were used, without data-driven feature selection. The 24 configurations were evaluated by 5-fold stratified cross-validation (Supplementary Table S1).

To compare families on equal terms, every family was then tuned with its best imbalance strategy by Bayesian optimisation (Optuna, tree-structured Parzen estimator [30]) on the same folds, with 40 trials for every family. Search spaces were deliberately wide, extending to shallow trees and stumps, slow learning rates and up to 2,000 boosting rounds (Supplementary Table S2). The stacking ensemble used the tuned base learners. Tuned hyperparameters were kept only if they improved on the defaults. The family with the highest tuned cross-validated AUROC (to three decimal places) was selected as the full model. Tuning and selection on the same folds make the internal estimates slightly optimistic, but do not affect the temporal estimates. The overall workflow is shown in Supplementary Figure S1.

2.10 Benchmarks

Six comparators were evaluated on the same temporal validation set, in addition to every tuned model family:

1. **OST** [5], $0.2 \times$ (weight in kg − age in years). Its discrimination and its commonly used cut-offs (< 2 and < -1) [5, 31] were evaluated without any fitting, overall and by sex. At each cut-off, we compared it with the full model at a threshold that referred the same proportion of participants, with a bootstrap CI for the number of additional cases detected. For calibration, net benefit and the Youden threshold, the score was converted to a probability by logistic recalibration on the development data ("recalibrated OST").
2. **ORAI** [7]: 15, 9 or 5 points for age ≥75, 65–74 or 55–64 years; 9 or 3 points for weight <60 or 60–69 kg; and 2 points for no estrogen use. NHANES records ever rather than current use of female hormones, so ever use was taken as estrogen use.
3. **SCORE** [8]: 5 points if not Black, 4 for rheumatoid arthritis, 4 for each fracture site (maximum 12), 3 times the first digit of age, and 1 for no estrogen use, minus weight in pounds divided by 10 (truncated). The original score counts non-traumatic wrist, rib and hip fractures after age 45; NHANES records hip, wrist and spine fractures of any cause, which we used instead. Rheumatoid arthritis was taken from the arthritis-type question. Rheumatoid arthritis was used only for SCORE.
4. **A three-variable logistic regression** of age, weight and sex.
5. **A Boruta + XGBoost pipeline** following the approach of Zhang et al. [10]. Boruta [32] selected features from our candidate set on the development data, and XGBoost, tuned with the same budget as the other families, was fitted to them. Because the original feature set and outcome definition differ, this tests the approach rather than the published model.
6. **The race-free model** (below).

ORAI and SCORE were developed in women; we applied them to both sexes, as we did OST. Like OST, both scores were evaluated as published for discrimination and converted to probabilities by logistic recalibration on the development data for calibration and net benefit. To mirror the published analysis, we also report all combinations of three feature-selection methods (Boruta, Lasso [33] and XGBoost importance) with four classifiers (LR, SVM, RF and XGBoost), evaluated by 10-fold cross-validation on the development set. Features were selected once on the full development set, so these estimates are optimistic and are reported only for comparison with the published study.

2.11 Recalibration and thresholds

Every model was recalibrated by Platt scaling [34], a logistic recalibration fitted on its out-of-fold development predictions. Internal calibration was assessed with cross-fitted recalibration, in which each fold was recalibrated using only the other folds. The primary classification threshold was set by the Youden index [35] on development predictions and applied unchanged to the temporal validation set. Because the Youden index weights sensitivity and specificity equally, which is not necessarily the right trade-off for screening, we also report a threshold giving 90% sensitivity in the development set and fixed risk thresholds of 3%, 5% and 10%.

2.12 Performance evaluation

Discrimination was assessed by AUROC and the area under the precision–recall curve (AUPRC). At each threshold we report sensitivity, specificity, positive and negative predictive values (PPV,

NPV), the proportion referred and the number of cases detected. Calibration was assessed by grouped and loess-smoothed calibration plots, the Brier score, the calibration slope and calibration-in-the-large [11]. Clinical utility was assessed by decision curve analysis [12] over thresholds of 0.01–0.40; at selected thresholds we report net benefit with bootstrap CIs, and true and false positives per 100 people.

Unless stated otherwise, 95% confidence intervals (CIs) and bootstrap P values were obtained from 1,000 bootstrap resamples of the validation set, with the same resamples used for every model. Differences in AUROC were estimated by paired bootstrap with a two-sided percentile P value; DeLong P values [36, 37] gave the same conclusions and are provided with the code.

2.13 Hypotheses and multiplicity

Comparisons of the full model with other models were organised as a hierarchy of three families, defined in this revision of the analysis:

1. **Primary:** the full model versus OST, the most widely used simple tool, tested at $\alpha = 0.05$ without adjustment.
2. **Secondary** (tested only if the primary comparison was significant): the full model versus ORAI, SCORE, the three-variable model, the Boruta + XGBoost pipeline and the race-free model, with Holm adjustment [38] within this family of five.
3. **Equivalence of ML and regression:** for each of the seven other tuned families, we asked whether its AUROC was equivalent to that of the full model within a margin of ± 0.02, chosen as a difference too small to matter for referral decisions. Equivalence was concluded if the 90% paired bootstrap CI of the difference lay entirely within the margin (two one-sided tests at $\alpha = 0.05$). Holm-adjusted two-sided P values for a difference are also reported.

The short form was compared descriptively. All other P values are unadjusted and exploratory.

2.14 Internal–external validation and model-fitting uncertainty

Because a single temporal split gives one estimate, we also performed internal–external cross-validation by survey cycle [39]: each of the five cycles was held out in turn, the full model (with the selected hyperparameters), the three-variable model and recalibrated OST were developed on the other four, and they were evaluated on the held-out cycle. AUROC differences in each cycle were estimated by paired bootstrap and pooled across cycles with a DerSimonian–Laird random-effects model [40], with a 95% prediction interval; single AUROCs were pooled using Hanley–McNeil standard errors [41].

The validation-set bootstrap does not reflect uncertainty in model fitting. We therefore also ran a refit bootstrap (200 replicates), in which the development set and the validation set were both resampled, the full model and the three-variable model were refitted to each development resample (hyperparameters fixed), and the AUROC difference with OST was computed on the validation resample.

2.15 Survey weighting and non-response

Models were fitted without survey weights, as is usual for prediction models. To estimate population-level performance, we also report AUROC, Brier score and observed versus predicted prevalence weighted by the 2017–2018 examination weights [42], with design-based 95% CIs from a Rao–Wu rescaled bootstrap over primary sampling units within strata (500 replicates). Because 22.5% of eligible participants had no valid DXA, we additionally multiplied the examination weights by the inverse of the probability of having a DXA result, estimated by logistic regression on age, sex, education, PIR, BMI (with a squared term and missing-value indicators), race/ethnicity and cycle among all participants aged 50 or older.

2.16 Incremental value of predictor groups

To show where the predictive information lies, we fitted nested LR models, adding predictor groups in clinical order: age, weight and sex; other body measures and socio-demographic factors; clinical risk factors and history; laboratory tests; and race/ethnicity. Each model was developed and validated as above, and each step is summarised by the paired bootstrap 95% CI and P value for the change in AUROC.

2.17 Subgroup and fairness analysis

In the temporal validation set, we evaluated performance by sex, age group (50–64, 65–79, $\geq$80 years) and race/ethnicity, with non-Hispanic Asian adults reported separately from other/multiracial adults. For each subgroup we report the number of events, AUROC and sensitivity (95% bootstrap CIs), specificity at the global threshold, and mean predicted risk versus observed prevalence. We also evaluated four clinical strata that correspond to screening recommendations [4]: women aged 50–64 years (screened if at increased risk), women aged 65 or older (screened routinely), men aged 50–69 and men aged 70 or older. In each we compared all benchmarks and the cases detected and referrals made by the full model at the global threshold and by OST < 2. We also examined two remedies: **sex-specific thresholds**, chosen by the Youden index within each sex on the development data; and **a race-free model**, with the same configuration but without the race/ethnicity predictors [14].

2.18 Short-form model

Predictor importance was ranked by mean absolute SHapley Additive exPlanations (SHAP) value [43] on 500 randomly selected development-set participants (permutation explainer with 100 background samples, probability scale before recalibration); all SHAP figures use this sample. The short form was restricted to predictors that a clinician can obtain at a visit without a blood test and that are not socially assigned: race/ethnicity, PIR and education were excluded, as was BMI (redundant with weight and height). The 10 most important remaining predictors were fitted with LR, developed and evaluated in the same way as the full model, overall and by sex, and are reported as an explicit equation.

2.19 Sensitivity analyses

We repeated the development and validation of the full model, OST and the three-variable model for the two spine-inclusive outcomes, and for the primary outcome with diagnosed or treated participants included. We also refitted the full model without non-Hispanic Asian participants, on 2013–2014 alone (same scanner type as 2017–2018), without the 2005–2006 cycle (which needed the most harmonisation), without laboratory predictors and without alcohol.

2.20 Software

Analyses used Python 3.9.6 with pandas 2.3.3, NumPy 2.0.2, scikit-learn 1.6.1 [44], imbalanced-learn 0.12.4, XGBoost 2.1.4, LightGBM 4.6.0, CatBoost 1.2.10, Boruta 0.4.3, Optuna 5.0.0, SHAP 0.49.1 and statsmodels 0.14.6, with a fixed random seed of 42. Two-sided $P < 0.05$ was considered significant.

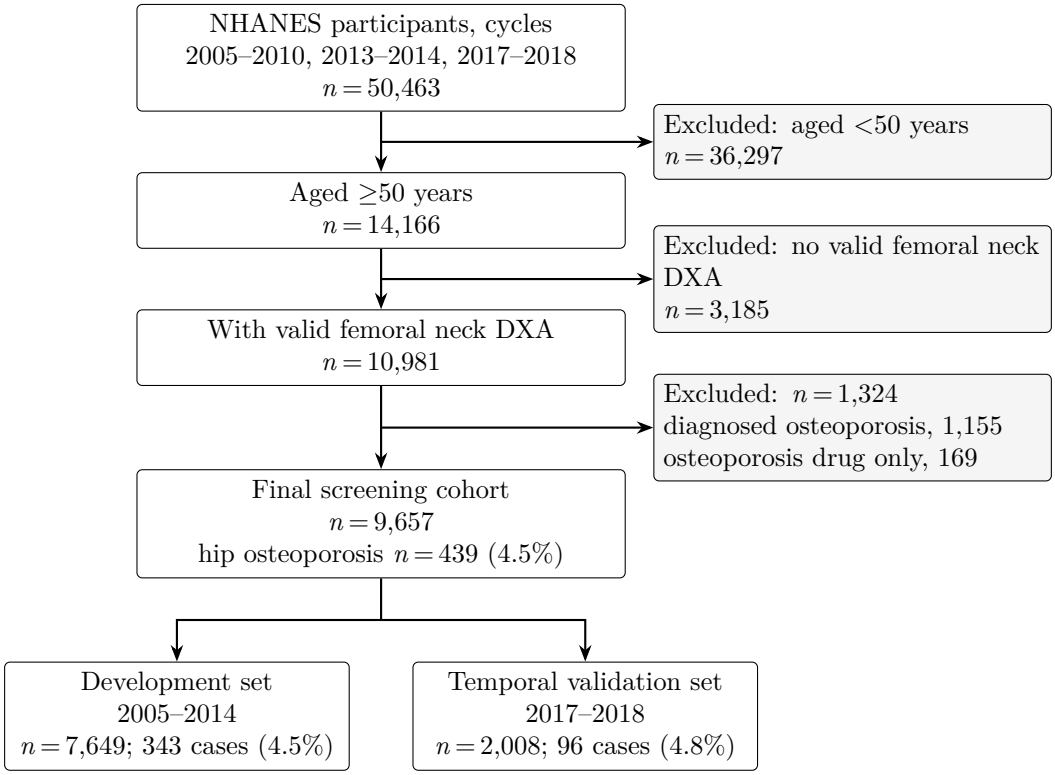


Figure 1: Participant flow. Percentages are the prevalence of osteoporosis at the hip (femoral neck or total hip T-score ≤ -2.5). No participant was excluded for missing more than 50% of predictors.

3 Results

3.1 Participant characteristics

The cohort comprised 9,657 adults (mean age 64.2 years; 43.0% women), of whom 439 (4.5%) had undiagnosed osteoporosis at the hip. The survey-weighted prevalence was 3.9%. Prevalence was stable across cycles (4.0%–4.8%), at 4.5% in the development set and 4.8% in the validation set. Compared with participants without osteoporosis, those with osteoporosis were older and

more often women and postmenopausal. They had lower body size and income, and more prior fractures (22.3% vs 11.6%) and strokes. They were less often non-Hispanic Black (Table 1).

The validation cohort differed from the development cohort in some respects (Supplementary Table S7): it had higher vitamin D and alkaline phosphatase, lower albumin and phosphorus, more diabetes, less hormone use and more participants of other or multiracial ethnicity (17.8% vs 5.9%). Participants without DXA were older (67.6 vs 64.2 years), heavier (BMI 33.0 vs 28.7 kg/m^2) and more often women and non-Hispanic Black (Supplementary Table S8).

Table 1: Baseline characteristics by osteoporosis status (n = 9,657). Values are mean (SD) or n (%); weighted columns are survey-weighted means or percentages. P values are from unweighted Welch's t-tests or χ^2 tests. Reproductive variables are expressed as a percentage of all participants (men coded 0).

	Unweighted			Weighted	
Characteristic	No (n = 9,218)	Yes (n = 439)	P	No	Yes
Age, years	63.88 (9.25)	71.14 (9.07)	<0.001	61.9	70.0
Female sex	3,833 (41.6)	318 (72.4)	<0.001	45.3	76.5
Postmenopausal	3,351 (37.1)	293 (70.4)	<0.001	39.7	74.7
Ever used female hormones	1,299 (14.4)	70 (16.9)	0.18	17.8	19.5
Prior hip, wrist or spine fracture	1,069 (11.6)	98 (22.3)	<0.001	13.4	23.4
Parental hip fracture	889 (9.9)	45 (10.7)	0.65	11.1	12.5
Ever daily glucocorticoid use	495 (5.4)	22 (5.1)	0.85	6.1	7.3
Alcohol ≥3 drinks/day	222 (2.5)	3 (0.7)	0.036	2.5	1.3
Current smoker	1,562 (17.0)	92 (21.0)	0.035	15.6	25.3
Stroke	512 (5.6)	45 (10.3)	<0.001	4.4	8.4
Diabetes	1,771 (19.2)	77 (17.5)	0.42	14.7	12.3
Non-Hispanic Black	1,991 (21.6)	43 (9.8)	<0.001	9.7	4.6
Income-to-poverty ratio	2.74 (1.62)	2.10 (1.37)	<0.001	3.33	2.43
BMI, kg/m^2	28.94 (5.53)	24.41 (4.46)	<0.001	28.97	24.37
Weight, kg	81.51 (17.75)	62.23 (13.56)	<0.001	82.8	62.4
Height, cm	167.67 (9.84)	159.44 (8.93)	<0.001	168.8	159.9
Waist, cm	101.60 (13.62)	90.73 (13.29)	<0.001	101.7	90.0
Alkaline phosphatase, U/L	74.31 (24.62)	82.85 (28.38)	<0.001	71.9	82.5
Uric acid, mg/dL	5.72 (1.43)	5.29 (1.44)	<0.001	5.62	5.12
25(OH) vitamin D, nmol/L	67.37 (27.85)	67.17 (30.44)	0.90	72.8	69.3

3.2 Re-implementation of the published approach

In 10-fold cross-validation, LR had the highest AUROC with every feature set (0.871–0.879). XGBoost (0.845–0.855) and RF (0.848–0.854) followed, and SVM with default settings performed worst (0.653–0.777) (Supplementary Table S6). Accuracy was 0.954–0.957 for all 12 combinations regardless of discrimination, almost identical to the 95.5% of the development set without osteoporosis (the accuracy of predicting "no osteoporosis" for everyone). Accuracy therefore did not separate the models. Boruta and XGBoost importance each selected 10 predictors, and Lasso retained 34 (Supplementary Table S5).

3.3 Model comparison and tuning

With default hyperparameters, cross-validated AUROC across the 24 configurations ranged from 0.785 (SVM without imbalance handling) to 0.876 (LR without imbalance handling) (Supplementary Table S1). Imbalance handling raised the AUROC of SVM (0.785 to 0.841 with class

weighting) and of RF (0.853 to 0.861 with class weighting or SMOTE), but not of the other families, and SMOTE lowered the AUPRC of every tree-based model. Class weighting and SMOTE raised the raw Brier score of LR from 0.035 to 0.127–0.137, reflecting inflated predicted risks; recalibration removed this difference.

Tuning narrowed the gaps between families (Table 2). With each family's best default configuration, the gap to LR ranged from 0.003 (LR-spline) to 0.035 (SVM). Tuning raised the AUROC of SVM by 0.035, of LightGBM by 0.023 and of XGBoost by 0.018, but LR by only 0.003. After tuning, seven of the eight families lay within 0.006 of each other (0.873–0.879), and RF reached 0.864. Penalised LR remained the highest (0.879; $C = 0.13$) and was selected as the full model. Gradient-boosting models converged on very shallow trees (XGBoost depth 1; LightGBM two leaves) with many rounds and slow learning rates, that is, on nearly additive models; these optima lie at the lower limit of what the model class allows, not at an arbitrary search bound (Supplementary Table S2).

Table 2: Cross-validated AUROC of each model family before and after equal-budget tuning (Optuna, 40 trials per family, development set, same 5 folds), with its best imbalance strategy. Stacking used the tuned base learners. Search spaces and selected values are in Supplementary Table S2.

Family	Strategy	Trials	Default AUROC	Tuned AUROC	Change
LR	None	40	0.876	0.879	0.003
SVM	Class wt	40	0.841	0.876	0.035
LR-spline	None	40	0.873	0.875	0.002
XGBoost	None	40	0.857	0.875	0.018
LightGBM	None	40	0.852	0.875	0.023
Stacking	Class wt	–	0.865	0.874	0.009
CatBoost	None	40	0.865	0.873	0.008
RF	Class wt	40	0.861	0.864	0.003

3.4 Temporal validation against benchmarks

On the 2017–2018 cycle, the full model (penalised LR, 35 predictors) achieved an AUROC of 0.869 (95% CI 0.831–0.904), close to its internal estimate (0.877, 0.857–0.896) (Table 3, Figure 2).

Primary comparison. The full model exceeded OST (0.841) by 0.028 (95% CI 0.007–0.052; $P = 0.014$). When uncertainty from model refitting was included by resampling the development set as well, the 95% CI widened to −0.004 to 0.052 (difference ≤ 0 in 4.5% of 200 replicates), and the CI for the full model's AUROC to 0.824–0.899.

Secondary comparisons. The full model clearly exceeded ORAI (0.791; difference 0.078, 0.047–0.115; Holm $P = 0.005$). Its differences from SCORE (0.841; 0.029, 0.003–0.055; unadjusted $P = 0.028$, Holm $P = 0.11$), the three-variable model (0.851; 0.018, −0.001 to 0.039; Holm $P = 0.21$), Boruta + XGBoost (0.856; 0.013, −0.007 to 0.030; Holm $P = 0.27$) and the race-free model (0.863; 0.006, −0.001 to 0.012; Holm $P = 0.27$) were not significant after adjustment.

ML versus regression. The tuned ML families ranged from 0.862 (RF) to 0.872 (stacking). For six of the seven, the 90% CI of the difference from LR lay within ±0.02, so they were equivalent to LR; for RF, the upper limit (0.020) touched the margin, so equivalence could not be concluded, although RF was not better (difference 0.007, 90% CI −0.008 to 0.020; Holm

$P = 1.00$).

The data are therefore compatible with no gain from 35 predictors over age, weight and sex, and with gains of up to about 0.04–0.05 in AUROC over OST or SCORE.

At the development-set Youden threshold (predicted risk ≥4.8%), the full model referred 23.8% of participants and detected 77 of 96 cases (sensitivity 0.802, specificity 0.791, NPV 0.988).

Table 3: Temporal validation performance (n = 2,008; 96 with osteoporosis), with 95% bootstrap CIs. Sensitivity, specificity and the proportion referred are at each model's development-set Youden threshold; for OST, ORAI and SCORE, values other than AUROC and AUPRC are for the recalibrated score. All ML families were tuned with the same budget. ΔAUROC, full model minus each alternative (paired bootstrap; 95% CI, except 90% CI for the equivalence family); P, two-sided bootstrap P; P_{Holm}, Holm-adjusted within its family. Equivalence margin ±0.02.

Model	AUROC (95% CI)	AUPRC	Brier	Sens.	Spec.	Referred	Slope (95% CI)	ΔAUROC (95% CI)	P	P_{Holm}
Full model (LR, 35 predictors)	**0.869 (0.831–0.904)**	**0.335**	**0.038**	**0.802**	**0.791**	**23.8%**	**1.00 (0.85–1.18)**	–	–	–
Primary comparison										
OST	0.841 (0.796–0.883)	0.280	0.040	0.823	0.717	30.9%	0.98 (0.79–1.20)	0.028 (0.007–0.052)	0.014	–
Secondary comparisons (Holm-adjusted within family)										
LR: age, weight, sex	0.851 (0.808–0.890)	0.312	0.039	0.781	0.783	24.4%	1.04 (0.86–1.28)	0.018 (−0.001 to 0.039)	0.070	0.21
SCORE	0.841 (0.799–0.879)	0.263	0.040	0.833	0.725	30.2%	1.04 (0.87–1.25)	0.029 (0.003–0.055)	0.028	0.11
ORAI	0.791 (0.742–0.834)	0.177	0.042	0.750	0.735	28.8%	0.93 (0.75–1.11)	0.078 (0.047–0.115)	0.001	0.005
Boruta + XGBoost pipeline	0.856 (0.820–0.892)	0.295	0.039	0.844	0.680	34.5%	1.04 (0.89–1.24)	0.013 (−0.007 to 0.030)	0.18	0.27
Full model without race	0.863 (0.825–0.898)	0.325	0.038	0.792	0.798	23.1%	1.00 (0.84–1.19)	0.006 (−0.001 to 0.012)	0.14	0.27
Tuned ML families vs LR: equivalence (90% CI, margin ±0.02; Holm-adjusted within family)										
LR-spline	0.869 (0.832–0.903)	0.320	0.038	0.802	0.793	23.6%	1.06 (0.91–1.25)	0.000 (−0.008 to 0.007)	0.94	1.00
SVM	0.870 (0.832–0.906)	0.367	0.037	0.740	0.835	19.2%	1.03 (0.87–1.22)	−0.001 (−0.010 to 0.009)	0.87	1.00
Random forest	0.862 (0.823–0.900)	0.313	0.038	0.812	0.742	28.4%	1.04 (0.87–1.24)	0.007 (−0.008 to 0.020)	0.40	1.00
XGBoost	0.868 (0.834–0.901)	0.320	0.038	0.771	0.828	20.0%	1.04 (0.90–1.22)	0.001 (−0.010 to 0.012)	0.88	1.00
LightGBM	0.867 (0.832–0.902)	0.325	0.038	0.802	0.745	28.1%	1.06 (0.92–1.26)	0.002 (−0.010 to 0.012)	0.83	1.00
CatBoost	0.867 (0.830–0.905)	0.333	0.038	0.823	0.741	28.6%	1.07 (0.92–1.26)	0.002 (−0.011 to 0.013)	0.86	1.00
Stacking	0.872 (0.836–0.908)	0.338	0.038	0.812	0.769	25.8%	1.04 (0.89–1.24)	−0.003 (−0.012 to 0.006)	0.51	1.00
Descriptive (not part of a comparison family)										
Short form (10 predictors, no blood test)	0.852 (0.810–0.887)	0.310	0.039	0.812	0.751	27.6%	0.96 (0.80–1.17)	0.018	0.018	–

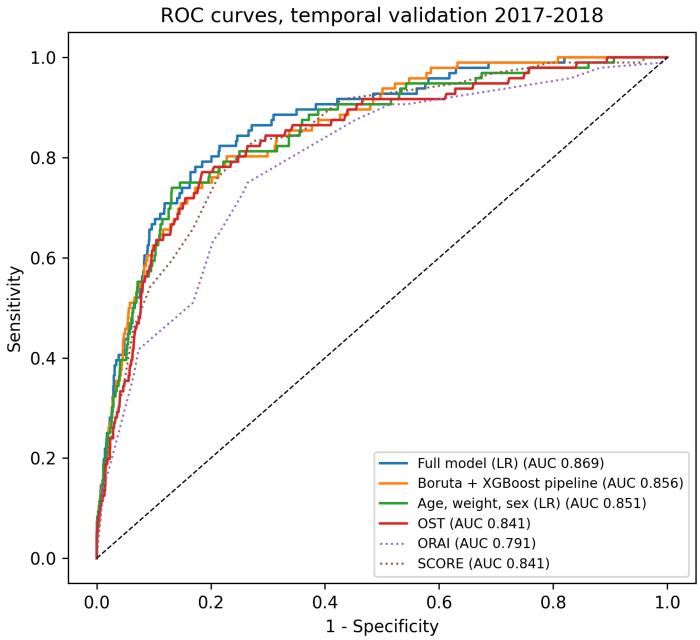


Figure 2: Receiver operating characteristic curves in the temporal validation set (2017–2018) for the full model and the five simple or published benchmarks (ORAI and SCORE dotted).

Weighted to the 2017–2018 survey population, AUROCs were slightly higher and ranked the models similarly: 0.889 (95% CI 0.835–0.937) for the full model, 0.870 for the three-variable

model, 0.868 for OST, 0.860 for SCORE and 0.819 for ORAI (Supplementary Table S10). After additional adjustment for DXA non-response (77.5% of eligible participants had a DXA result), the AUROC of the full model was 0.872 (0.812–0.934), against 0.856 for the three-variable model and 0.849 for OST; its difference from OST was 0.023 (−0.003 to 0.051). Weighted observed and predicted prevalence were 4.2% and 4.1%. Because the adjustment uses only a few observed characteristics, these estimates indicate, rather than establish, performance in the US population aged 50 or older.

3.5 Internal–external validation across cycles

When each cycle was held out in turn, the full model's AUROC ranged from 0.841 (2013–2014) to 0.917 (2005–2006) and exceeded both OST and the three-variable model in every cycle (Table 4). The pooled gain over OST was 0.025 (95% CI 0.015–0.034; 95% prediction interval 0.009–0.041) and over the three-variable model 0.019 (0.010–0.028; prediction interval 0.004–0.034), with little heterogeneity between cycles (I^2 0% and 3%). Calibration slopes ranged from 0.81 to 1.24. These estimates use far more events (439) than the single temporal split, and agree with it in direction and size.

Table 4: Internal–external cross-validation: each cycle held out in turn, with models developed on the other four cycles. Δ, full model minus comparator (paired bootstrap 95% CI). Pooled estimates from a DerSimonian–Laird random-effects model; single AUROCs pooled with Hanley–McNeil standard errors. The 2017–2018 row is the main temporal validation.

			AUROC					
Held-out cycle	*n*	Cases	Full model	Age, weight, sex	OST	Δ vs OST (95% CI)	Δ vs age, weight, sex (95% CI)	Slope
2005–2006	1,422	65	0.917	0.910	0.900	0.017 (−0.003 to 0.040)	0.007 (−0.008 to 0.022)	1.11
2007–2008	2,090	84	0.900	0.873	0.867	0.034 (0.010–0.058)	0.028 (0.006–0.049)	1.24
2009–2010	2,164	100	0.878	0.853	0.858	0.020 (0.003–0.038)	0.025 (0.008–0.040)	0.93
2013–2014	1,973	94	0.841	0.811	0.808	0.033 (0.006–0.063)	0.029 (0.005–0.058)	0.81
2017–2018	2,008	96	0.869	0.851	0.841	0.028 (0.007–0.052)	0.018 (−0.001 to 0.039)	1.00
Pooled (random effects)	9,657	439	0.882	0.861	0.856	0.025 (0.015–0.034)	0.019 (0.010–0.028)	
95% prediction interval			0.817–0.947			0.009–0.041	0.004–0.034	

3.6 Comparison with OST cut-offs

At OST's commonly used cut-offs, the full model detected slightly more cases at the same referral rate, with similar specificity (Table 5). At the referral rate of OST < 2 (39.1% of participants), it detected 86 rather than 83 of the 96 cases (3 more, 95% CI −2 to 8); at the referral rate of OST < -1 (14.9%), 68 rather than 62 (6 more, −2 to 14). By sex, the full model exceeded OST by 0.019 (−0.005 to 0.046) in women and 0.020 (−0.031 to 0.068) in men (Supplementary Table S11). In men, OST < -1 detected 12 of 24 cases (sensitivity 0.500) while referring 11.2%, and OST < 2 detected 20 of 24 (0.833) while referring 32.2%.

3.7 Operating points

The choice of threshold mattered more than the choice of model (Supplementary Table S12). A threshold set for 90% sensitivity in the development set (predicted risk 2.3%) detected 86 of 96 cases (0.896) in validation while referring 39.3% of participants, close to the referral rate of OST

Table 5: OST at its published cut-offs (no fitting) versus the full model at a threshold referring the same proportion of participants (temporal validation set, 96 cases). Additional cases: full model minus OST, with 95% bootstrap CI.

Referral rate	Rule	Cases detected	Sensitivity	Specificity	Additional cases
39.1%	OST < 2	83	0.865	0.633	
	Full model	86	0.896	0.634	3 (−2 to 8)
14.9%	OST < -1	62	0.646	0.876	
	Full model	68	0.708	0.879	6 (−2 to 14)

< 2 (39.1%), which detected 83. The Youden threshold (4.8%) referred 23.8% and detected 77 cases, and a 10% threshold referred 12.0% with a PPV of 0.261.

3.8 Calibration and clinical utility

The full model was well calibrated internally with cross-fitted recalibration (slope 0.99, intercept 0.00) and in temporal validation (slope 1.00, 95% CI 0.85–1.18; Brier score 0.038) (Figure 3). Predictions above 0.35 (1.7% of participants) lie outside the plotted range. The model slightly underestimated risk in 2017–2018 (calibration-in-the-large 0.09, 95% CI −0.17 to 0.29), as did recalibrated OST (0.15), ORAI (0.13), SCORE (0.13) and the three-variable model (0.16); none of these intercepts differed significantly from 0.

The full model, the three-variable model and Boruta + XGBoost had higher net benefit than referring everyone or no one at every threshold from 0.01 to 0.40; recalibrated OST did not at a threshold of 0.01 (Figure 4). ORAI had the lowest net benefit of all models at each tabulated threshold (0.05–0.20). Differences between models were small relative to their uncertainty (Supplementary Table S13). At a threshold of 0.10, net benefit was 0.022 (95% CI 0.015–0.030) for the full model and 0.018 (0.011–0.026) for recalibrated OST; per 100 people screened, the full model identified 3.14 true positives with 8.86 false positives, and OST 2.84 and 9.11.

3.9 Incremental value of predictor groups

Age, weight and sex alone gave a temporal-validation AUROC of 0.851 (Table 6). No further group of predictors produced a change whose 95% CI excluded zero; the largest single gain was 0.007, from other body measures and socio-demographic factors. All 32 predictors added to age, weight and sex together raised the AUROC by 0.018.

Table 6: Incremental value of predictor groups in nested logistic models. CV, 5-fold cross-validation in the development set; temporal, 2017–2018 validation set. ΔAUROC compares each model with the previous row (paired bootstrap 95% CI and P).

Predictors	k	CV AUROC	Temporal AUROC (95% CI)	AUPRC	ΔAUROC (95% CI)	P
Age, weight, sex	3	0.856	0.851 (0.808–0.890)	0.312	–	–
+ other body measures, socio-demographics	8	0.864	0.858 (0.816–0.894)	0.332	0.007 (−0.005 to 0.018)	0.21
+ clinical risk factors and history	21	0.869	0.859 (0.819–0.894)	0.318	0.001 (−0.010 to 0.012)	0.85
+ laboratory tests	31	0.872	0.863 (0.825–0.898)	0.325	0.005 (−0.002 to 0.011)	0.17
+ race/ethnicity	35	0.877	0.869 (0.831–0.904)	0.335	0.006 (−0.001 to 0.012)	0.14

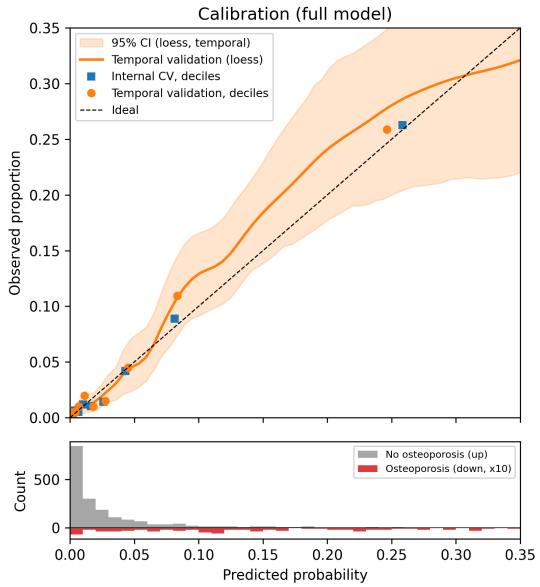


Figure 3: Calibration of the full model: deciles of predicted risk internally (cross-fitted) and in temporal validation, and a loess curve for the validation set with a 95% bootstrap band (200 replicates); dashed line, perfect calibration. The axes stop at 0.35 (98th percentile of predictions). Lower panel: predicted risks of participants without (up) and with (down, counts ×10) osteoporosis.

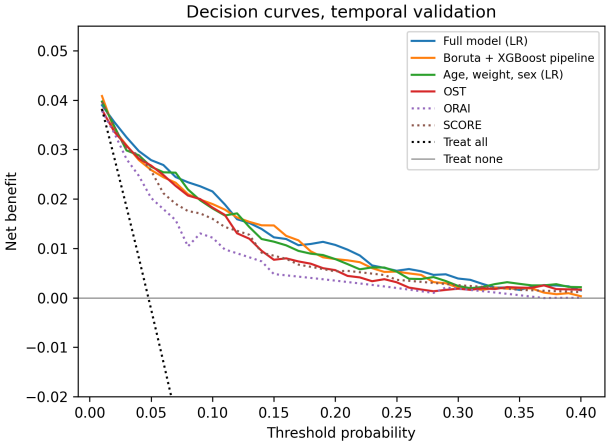


Figure 4: Decision curves in the temporal validation set (OST, ORAI and SCORE recalibrated to probabilities).

3.10 Subgroup performance and fairness

At the global threshold, sensitivity was 0.847 in women (61 of 72 cases) but 0.667 in men (16 of 24) (Table 7). Discrimination within each sex was similar (AUROC 0.853 and 0.820, with overlapping CIs). Sensitivity rose with age, from 0.656 at 50–64 years to 0.931 at 80 years or older, and adults aged 80 or older were slightly under-predicted (mean predicted risk 12.9% vs observed 14.5%).

The two remedies had modest effects:

- **Sex-specific thresholds** were 6.7% for women and 3.5% for men, against a global threshold of 4.8%. For men, sensitivity rose from 0.667 to 0.750 (18 of 24 cases) and referrals from 13.6% to 17.9%. For women, referrals fell from 37.4% to 29.4%, and sensitivity from 0.847 to 0.819 (59 of 72 cases). Overall, sex-specific thresholds detected the same 77 cases as the global threshold while referring fewer participants (458 vs 477; 22.8% vs 23.8%), shifting two detected cases from women to men.
- **Removing race** reduced the AUROC slightly (0.863 vs 0.869). It shifted predicted risk towards the pooled average: risk was over-predicted in non-Hispanic Black adults (3.2% predicted vs 1.4% observed, against 2.1% with race) and under-predicted in non-Hispanic white adults (5.5% vs 7.1%, against 6.3%).

Non-Hispanic Asian adults had a high prevalence (7.3%, 20 cases) and the model discriminated well in them (AUROC 0.904, 0.840–0.955; sensitivity 0.900), although it under-predicted their risk slightly (6.0%); the model refitted without them performed similarly (0.861; Supplementary

Table S14). The other racial and ethnic subgroups had 4–8 cases each, and their sensitivity estimates were very imprecise (for example, 0.571, 95% CI 0.143–1.000, from 7 cases in non-Hispanic Black adults), so no conclusions about differences in detection between these groups can be drawn.

Table 7: Performance of the full model by subgroup in the temporal validation set, at the global threshold (4.8%). Pred., mean predicted risk. 95% bootstrap CIs. The last three columns show the race-free model at its own threshold (5.2%). In the models, non-Hispanic Asian adults are part of the other/multiracial category (see Methods).

				Full model				Race-free model		
Subgroup	n	Cases	Prev.	Pred.	AUROC (95% CI)	Sensitivity (95% CI)	Spec.	Pred.	AUROC	Sens.
Women	858	72	8.4%	7.2%	0.853 (0.811–0.894)	0.847 (0.754–0.922)	0.669	7.3%	0.847	0.847
Men	1,150	24	2.1%	2.4%	0.820 (0.733–0.899)	0.667 (0.468–0.850)	0.876	2.5%	0.816	0.625
Age 50–64	1,157	32	2.8%	2.7%	0.861 (0.790–0.924)	0.656 (0.500–0.818)	0.864	2.7%	0.849	0.625
Age 65–79	651	35	5.4%	5.0%	0.855 (0.787–0.912)	0.829 (0.696–0.944)	0.752	5.2%	0.847	0.829
Age ≥80	200	29	14.5%	12.9%	0.793 (0.696–0.882)	0.931 (0.833–1.000)	0.450	13.0%	0.800	0.931
Non-Hispanic white	718	51	7.1%	6.3%	0.851 (0.799–0.897)	0.824 (0.717–0.917)	0.736	5.5%	0.855	0.804
Non-Hispanic Black	485	7	1.4%	2.1%	0.782 (0.570–0.930)	0.571 (0.143–1.000)	0.874	3.2%	0.776	0.571
Mexican American	231	8	3.5%	2.7%	0.816 (0.674–0.956)	0.500 (0.143–0.889)	0.883	3.7%	0.803	0.625
Other Hispanic	217	6	2.8%	3.8%	0.963 (0.930–0.987)	1.000 (1.000–1.000)	0.806	3.8%	0.957	1.000
Non-Hispanic Asian	275	20	7.3%	6.0%	0.904 (0.840–0.953)	0.900 (0.750–1.000)	0.698	5.8%	0.904	0.850
Other/multiracial (non-Asian)	82	4	4.9%	3.9%	0.821 (0.450–0.988)	0.750 (0.000–1.000)	0.744	3.8%	0.814	0.750

3.11 Clinical strata

In women aged 65 or older, the group for whom screening is routinely recommended (45 cases), the full model's AUROC was 0.836, against 0.811 for OST, 0.811 for the three-variable model, 0.807 for SCORE and 0.783 for ORAI (Table 8); the difference from OST was 0.025 (−0.019 to 0.068). Because prevalence is high in this group, OST < 2 referred 71.1% of these women to detect 42 cases, whereas the full model at its global threshold referred 55.0% and detected 43. In women aged 50–64 (27 cases), SCORE had the highest AUROC (0.838 vs 0.818 for the full model). In men aged 50–69, the full model exceeded OST by 0.080 (0.013–0.152), but this estimate rests on only 9 cases; in men aged 70 or older (15 cases), the models were similar, and OST < 2 detected more cases than the full model at its global threshold (14 vs 10) by referring twice as many men (59.8% vs 30.3%). None of these stratum-specific comparisons was part of the testing hierarchy.

Table 8: Performance in clinical strata of the temporal validation set. AUROC of each model; Δ, full model minus OST (paired bootstrap 95% CI). The last two columns give cases detected and the proportion referred by the full model at its global Youden threshold (4.8%) and by OST < 2 (no fitting). Exploratory; not part of the testing hierarchy.

			AUROC						Detected, referred	
Stratum	n	Cases	Full	Age, weight, sex	OST	SCORE	ORAI	Δ vs OST (95% CI)	Full model	OST < 2
Women 50–64 years	547	27	0.818	0.804	0.804	0.838	0.749	0.014 (−0.030 to 0.065)	18/27, 27.4%	21/27, 35.5%
Women ≥65 years	311	45	0.836	0.811	0.811	0.807	0.783	0.025 (−0.019 to 0.068)	43/45, 55.0%	42/45, 71.1%
Men 50–69 years	787	9	0.813	0.733	0.734	0.751	0.787	0.080 (0.013–0.152)	6/9, 5.8%	6/9, 19.4%
Men ≥70 years	363	15	0.741	0.732	0.733	0.756	0.629	0.008 (−0.074 to 0.100)	10/15, 30.3%	14/15, 59.8%

3.12 Predictor importance and short form

In the development sample, the most important predictors by mean absolute SHAP value were the body-size measures (BMI, weight, height and waist), postmenopausal status and age (Figure 5; Supplementary Table S3), followed by hormone use, non-Hispanic Black ethnicity, PIR and prior fracture; no laboratory test ranked in the top ten. Dependence and individual-level plots are shown in Supplementary Figures S2 and S3. Because the body-size measures are strongly correlated, their individual contributions should be interpreted together.

The short form used weight, postmenopausal status, age, height, waist circumference, female hormone use, prior fracture, arthritis, cancer and stroke, all available at a clinic visit without a blood test. It achieved an AUROC of 0.852 (0.810–0.887; calibration slope 0.96), lower than the full model (difference 0.018; $P = 0.018$). It performed similarly to the three-variable model (difference 0.000, −0.013 to 0.014) and was not significantly better than OST (0.010, −0.007 to 0.028). At its development threshold (4.0%) it referred 27.6% of participants and detected 78 of 96 cases. Its AUROC was 0.835 in women and 0.800 in men, in each case within 0.001 of the three-variable model and OST in that sex; its sensitivity was 0.889 in women and 0.583 in men (14 of 24 cases) (Supplementary Table S4). Table 9 gives its equation. The negative coefficients for arthritis and cancer reflect associations conditional on body size and age in these data, and should not be interpreted causally.

Table 9: Short-form equation (penalised logistic regression, recalibrated). Predicted probability = $1/(1 + e^{-\eta})$, where η is the intercept plus the sum of each coefficient multiplied by the predictor value. Units: weight in kg, age in years (capped at 80), height and waist in cm; binary predictors 1 = yes, 0 = no (postmenopausal and hormone use are 0 for men). Coefficients of correlated body-size measures are not individually interpretable. Development-set Youden threshold 4.0%.

Term	Coefficient	Odds ratio per unit
Intercept	−3.43385	
Weight	−0.09585	0.909
Postmenopausal	0.97643	2.655
Age	0.07253	1.075
Height	−0.00161	0.998
Waist	0.02118	1.021
Ever used female hormones	−0.83012	0.436
Prior hip, wrist or spine fracture	1.08190	2.950
Arthritis	−0.22820	0.796
Cancer	−0.30003	0.741
Stroke	0.69766	2.009

3.13 Sensitivity analyses

The conclusions did not change in any sensitivity analysis (Supplementary Table S14). Including the lumbar spine raised prevalence in the development set to 7.2% (complete scans only) and 8.4% (spine from at least two vertebrae), and in the validation set to 6.7% and 8.2%. For these outcomes the full model achieved AUROCs of 0.846 and 0.840, against 0.823 and 0.801 for OST and 0.836 and 0.824 for the three-variable model. With diagnosed or treated participants included, the AUROCs were 0.859, 0.824 and 0.839. Refitting without non-Hispanic Asian participants (0.861), on 2013–2014 alone (0.854), without the 2005–2006 cycle (0.869),

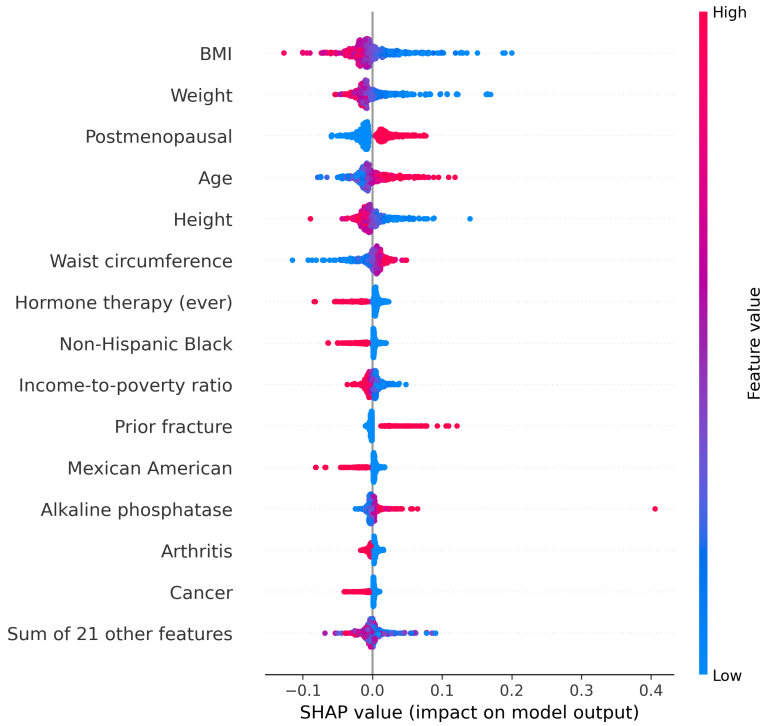


Figure 5: SHAP beeswarm summary for the full model in 500 development-set participants (the sample used for importance ranking). Colour indicates the predictor value; horizontal position indicates its effect on predicted risk (probability scale, before recalibration).

without laboratory predictors (0.863) or without alcohol (0.869) made little difference. Across the versions of the analysis that included these benchmarks (versions 2–6), the gain of the best model over OST was between 0.028 and 0.045, and over the three-variable model between 0.017 and 0.030 (Supplementary Table S16).

4 Discussion

4.1 Principal findings

In 9,657 older US adults neither diagnosed with nor treated for osteoporosis, a penalised logistic regression using 35 history, examination and laboratory predictors identified hip osteoporosis with an AUROC of 0.869 in a later survey cycle, with good calibration. When every model family was tuned with the same budget, SVM, random forest, three gradient-boosting methods, stacking and spline regression all performed within 0.01 of logistic regression, and six of the seven were formally equivalent to it. The full model exceeded OST by 0.028 in the temporal validation and by 0.025 when pooled across five held-out cycles, but the temporal CI included zero once uncertainty from refitting was considered. It clearly exceeded ORAI, but not SCORE or a three-variable model of age, weight and sex after adjustment. A short form using only clinically obtainable, race-free predictors performed like age, weight and sex. A single threshold detected two-thirds of cases in men.

4.2 How much do additional predictors and machine learning add?

Most of the discriminative information was captured by age, weight and sex. OST, which uses only two of these, reached an AUROC of 0.841 without any model fitting, as did SCORE; ORAI, which groups age and weight into coarse bands, performed worse (0.791). No further group of predictors made a clear difference on its own, and together they added 0.018. This agrees with evidence that, for tabular clinical data with modest numbers of predictors, ML rarely outperforms logistic regression [15]. The relationships here are largely monotonic and additive, so flexible models have little additional structure to exploit; spline terms added nothing either.

Equal tuning was important for this conclusion. With default settings, the best configuration of each ML family trailed LR by up to 0.035 in cross-validated AUROC (SVM), which could have been misread as evidence that ML is worse; with tuning, all families converged. Tuned boosting models chose stumps or two-leaf trees, effectively additive models, which is further evidence that these data contain little interaction structure. Conversely, comparing a tuned ML model with an untuned regression could exaggerate the value of ML.

Our analyses do not exclude modest gains, and they bound them. Over OST, the temporal validation is compatible with differences in AUROC from about 0 to 0.05, and the internal–external validation, which uses all 439 cases, gives a narrower range of 0.015–0.034. At OST's referral rates this corresponds to detecting a few more of the 96 cases (3 and 6 more in our data, with CIs that include 2 fewer). Whether such a gain justifies a tool that needs 35 inputs, including blood tests, depends on the setting; in most settings, we think it would not. Among the simple indices, OST and SCORE performed similarly overall, and SCORE did best in women aged 50–64, but it needs rheumatoid arthritis and fracture history, which are less reliably recorded than age and weight. We did not evaluate FRAX calculated without BMD, which estimates fracture probability rather than low BMD.

4.3 Comparison with the published study

Zhang et al. reported their results mainly as accuracy [10]. In our re-implementation, all 12 feature-selection and classifier combinations had an accuracy of 0.954–0.957, including SVMs with AUROC as low as 0.653, each close to the no-information rate of 95.5%. Their outcome definition and population differ from ours, so their figures cannot be compared directly with ours, but our results show that accuracy cannot rank osteoporosis models when the condition is uncommon. On temporal validation, a tuned Boruta + XGBoost pipeline (0.856) performed similarly to the three-variable model, so feature selection followed by a flexible learner added complexity without measurable benefit.

4.4 Class imbalance and calibration

Class-imbalance corrections improved discrimination only for untuned SVM and, slightly, for random forest, and SMOTE lowered the AUPRC of every tree-based model. Class weighting and resampling inflated predicted risks, roughly quadrupling the Brier score of LR until recalibration removed the difference. This is consistent with van den Goorbergh et al. [16]. When valid absolute risks are needed, models should be fitted on the original class distribution, with a

decision threshold chosen for the intended use.

4.5 Thresholds

The Youden threshold is a statistical convention, not a clinical choice. For screening, where missing a case usually costs more than an unnecessary DXA, a lower threshold is often preferable. In our data, a threshold chosen for 90% sensitivity in development transferred well to validation (0.896) and referred 39% of participants, as many as OST < 2, detecting 86 cases against 83 for OST. The operating point should be chosen locally, from the referral capacity and the relative costs of missed and unnecessary scans, and Table S12 and the decision curves provide the information to do so.

4.6 Subgroups and fairness

The model ranked men about as well as women (AUROC 0.820 vs 0.853), but a single threshold detected only 16 of 24 cases in men, reflecting their low baseline risk. Sex-specific Youden thresholds raised sensitivity in men to 0.750 and lowered referrals in women, detecting the same total number of cases with fewer referrals; raising detection in men further would require a deliberately lower threshold and more referrals, a policy choice about the value of finding osteoporosis in men, who are already under-screened [4]. OST behaved similarly: at its stricter cut-off it detected only half of the cases in men.

In women aged 65 or older, for whom DXA is recommended regardless of risk scores [4], a tool is useful mainly for ordering patients when capacity is limited; there the full model detected as many cases as OST < 2 with fewer referrals, but with 45 cases this is a hypothesis rather than a finding. The strata where a tool would decide who is scanned at all, younger postmenopausal women and men, had 9–27 cases each, too few to choose between tools.

Removing race/ethnicity changed discrimination little but miscalibrated risk within groups: it over-predicted risk in non-Hispanic Black adults, whose observed prevalence was low, and under-predicted it in white adults. This illustrates the trade-off between calibration within groups and race-free prediction [13, 14]. The other/multiracial category changed composition over time, as non-Hispanic Asian adults were oversampled from 2011; the model nevertheless discriminated well in Asian adults, and results without them were similar. With only 4–8 cases in each other minority group, our data cannot show whether either model detects osteoporosis less well in a particular group.

4.7 Implementation considerations

For a tool to be used in practice, its inputs must be available where decisions are made. Age, sex, weight and height are recorded at almost every primary-care visit and are routinely stored in electronic health records (EHRs), which is why OST and the three-variable model can be computed automatically for every eligible patient. The short form adds waist circumference, menopausal status, hormone use, prior fracture and three common diagnoses. These are less consistently coded in EHRs, and in our data they added nothing measurable. The full model

also needs a biochemistry panel and vitamin D, which are available only for patients who happen to have been tested, and PIR, which is rarely recorded at all.

Any of these models can be implemented as a single linear equation in an EHR or a spreadsheet, at negligible cost (Table 9, Supplementary Table S15). Gradient-boosting models cannot be written in this way, and would need software integration and version control for no gain in performance. The main cost of screening lies in the DXA scans triggered, not in the tool, so the choice of threshold has far larger resource implications than the choice of model. For health systems that already use OST, our results suggest that replacing it with an ML model would bring little benefit, whereas choosing an appropriate threshold, and applying it to men as well as women, could matter more.

4.8 Strengths and limitations

This study has several strengths:

- a large sample with standardised DXA from a national probability survey, restricted to the intended screening population of undiagnosed, untreated adults;
- a hip-based outcome with stable prevalence across cycles, and spine-inclusive and scanner-restricted sensitivity analyses;
- a temporal validation cycle, supported by internal–external validation across all five cycles and a bootstrap including model refitting;
- equal tuning of every model family over wide search spaces, and direct comparison with three established screening indices, including OST's cut-offs overall, by sex and in guideline-defined strata;
- leakage-safe imputation, resampling and predictor selection;
- assessment of discrimination, calibration and net benefit with confidence intervals, an explicit testing hierarchy with equivalence testing, survey weighting and non-response adjustment;
- explicit reporting of subgroup event counts and uncertainty;
- published model equations, a record of every analysis version, and full reproducibility from public data and open code.

The study also has limitations:

1. **Cross-sectional design.** The model detects existing osteoporosis; it does not predict fractures. We did not compare it with FRAX, which estimates fracture probability rather than BMD.
2. **Analysis revisions.** The outcome definition and the cohort were revised, and the testing hierarchy, the ORAI and SCORE benchmarks and the internal–external validation were added, after the validation data had been examined in earlier versions of this analysis. All revisions were made for stated methodological reasons, and the results of every version are reported (Supplementary Table S16), but the temporal validation is not strictly untouched and the hierarchy was not prespecified.
3. **Hip-only primary outcome.** Excluding the spine underestimates osteoporosis as defined clinically, where the lowest T-score at any site is used. NCHS reported an age-adjusted prevalence of 12.6% at the femoral neck or lumbar spine in 2017–2018 [2], including people already diagnosed; our figures were lower because of the hip-only definition and the exclusion

of diagnosed and treated participants.

4. **Cross-cycle measurement changes.** DXA scanners, laboratory methods and the alcohol questionnaire changed between periods, and mean femoral neck BMD in women was about 0.15 SD lower after the scanner change. Results without laboratory predictors or alcohol, and with development restricted to the same scanner type, were similar.
5. **Self-reported information.** Risk factors and prior diagnosis were self-reported; a missing answer to the diagnosis question was treated as not diagnosed; prior fractures were of any cause; glucocorticoid use was recorded as ever rather than current use; and postmenopausal status was approximated from the absence of regular periods, which also captures hysterectomy and is nearly collinear with sex in older adults.
6. **Comparator re-implementation.** The Boruta + XGBoost pipeline was applied to our candidate predictors and outcome, because the published feature set was unavailable. ORAI and SCORE were derived in women and were applied here to both sexes; ORAI used ever rather than current estrogen use, and SCORE used fractures of any cause at the hip, wrist and spine rather than non-traumatic wrist, rib and hip fractures, which may understate their performance.
7. **Survey design.** Models were fitted without survey weights. Our non-response adjustment used only a few observed characteristics, and participants without DXA were heavier and more often women and non-Hispanic Black, so the cohort may not fully represent the US population aged 50 or older.
8. **Uncertainty.** With 96 validation events, differences of a few hundredths in AUROC cannot be excluded or confirmed from the temporal split alone. Most CIs reflect sampling of the validation set only; the refit bootstrap, which also reflects model fitting, kept hyperparameters fixed and used 200 replicates. Internal estimates are slightly optimistic because tuning and selection used the same folds. Men, younger women and minority racial and ethnic groups had few cases, limiting subgroup inference.
9. **US population only.** External validation is needed, including in settings where DXA access is most limited.

5 Conclusion

In older US adults neither diagnosed with nor treated for osteoporosis, a penalised logistic model using 35 clinical and laboratory predictors identified hip osteoporosis with good, temporally stable discrimination and calibration. Once tuned equally, seven ML and ensemble methods performed no better, and most were equivalent. Thirty-five predictors added about 0.02–0.03 in AUROC to OST and to age, weight and sex, a consistent but small gain, and a short form using only clinically obtainable predictors performed like age, weight and sex. For osteoporosis case-finding, simple tools capture most of the achievable discrimination; the threshold, and whether it is applied to men, matter more than the algorithm, and new models should be judged against these simple tools.

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Supplementary material

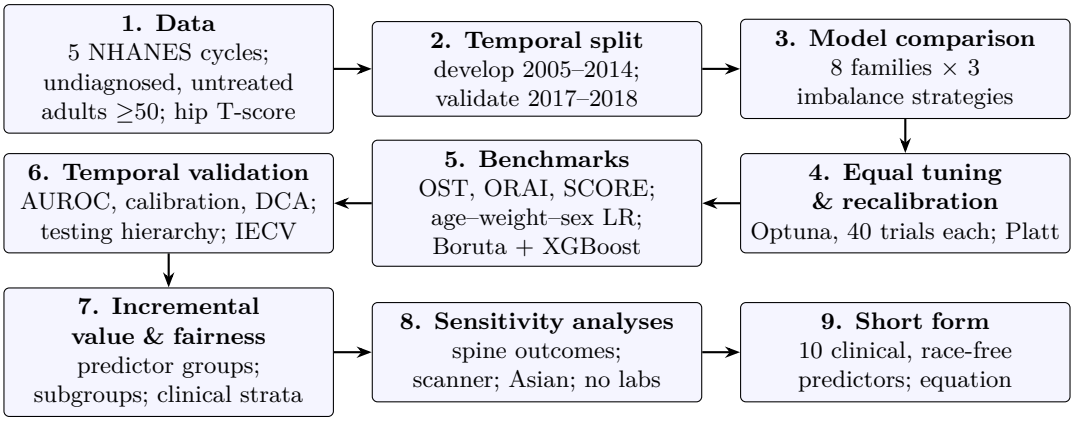


Figure S1: Study workflow. Imputation, scaling and resampling were fitted inside cross-validation folds; the 2017–2018 cycle was used only for evaluation.

Table S1: Cross-validated performance of 24 model × imbalance-strategy configurations with default hyperparameters (all 35 predictors, 5-fold CV, development set). Brier scores are shown before (raw) and after cross-fitted Platt recalibration (recal.). The selected configuration is in bold.

				Brier						Brier	
Model	Strategy	AUROC	AUPRC	raw	recal.	Model	Strategy	AUROC	AUPRC	raw	recal.
LR	**None**	**0.876**	**0.346**	**0.035**	**0.035**	XGBoost	None	0.857	0.275	0.038	0.037
LR	Class wt	0.872	0.346	0.137	0.035	XGBoost	Class wt	0.848	0.262	0.058	0.037
LR	SMOTE	0.871	0.336	0.127	0.035	XGBoost	SMOTE	0.855	0.266	0.041	0.037
LR-spline	None	0.873	0.339	0.035	0.035	LightGBM	None	0.852	0.275	0.039	0.037
LR-spline	Class wt	0.870	0.337	0.133	0.035	LightGBM	Class wt	0.842	0.258	0.050	0.038
LR-spline	SMOTE	0.868	0.331	0.125	0.036	LightGBM	SMOTE	0.848	0.267	0.040	0.037
SVM	None	0.785	0.239	0.038	0.038	CatBoost	None	0.865	0.294	0.037	0.037
SVM	Class wt	0.841	0.238	0.038	0.038	CatBoost	Class wt	0.851	0.299	0.072	0.037
SVM	SMOTE	0.812	0.212	0.055	0.039	CatBoost	SMOTE	0.841	0.246	0.042	0.038
RF	None	0.853	0.298	0.037	0.037	Stacking	None	0.863	0.302	0.037	0.038
RF	Class wt	0.861	0.303	0.043	0.036	Stacking	Class wt	0.865	0.311	0.036	0.036
RF	SMOTE	0.861	0.250	0.058	0.038	Stacking	SMOTE	0.864	0.270	0.045	0.038

Table S2: Hyperparameter search spaces and selected values (Optuna, tree-structured Parzen estimator, 40 trials per family, 5-fold CV AUROC in the development set). *Selected value within 5% of a search bound (3 of the tuned hyperparameters). For XGBoost depth 1 and LightGBM two leaves, the bound is the smallest possible tree (a stump); LightGBM's minimum child size of 198 lies at the upper bound and indicates strong regularisation.

Family	Hyperparameter	Search range (scale)	Selected
LR	C	0.001–100 (log)	0.135
LR-spline	C	0.001–100 (log)	1.92
SVM	C	0.01–100 (log)	1.05
	gamma	1.00e-05–1 (log)	0.00315
RF	n estimators	200–1000 (integer, step 100)	700
	max depth	3–20 (integer, step 1)	12
	min samples leaf	1–60 (integer, step 1)	27
	max features	0.05–1 (linear)	0.0934
XGBoost	n estimators	100–2000 (integer, step 50)	1600
	learning rate	0.002–0.3 (log)	0.0234
	max depth	1–8 (integer, step 1)	1*
	subsample	0.5–1 (linear)	0.706
	colsample bytree	0.5–1 (linear)	0.648
	min child weight	1–20 (log)	1.4
	reg lambda	0.001–10 (log)	2.9
LightGBM	n estimators	100–2000 (integer, step 50)	1300
	learning rate	0.002–0.3 (log)	0.0165
	num leaves	2–64 (integer, step 1)	2*
	min child samples	1–200 (integer, step 1)	198*
	subsample	0.5–1 (linear)	0.833
	colsample bytree	0.5–1 (linear)	0.614
	reg lambda	0.001–10 (log)	0.168
CatBoost	iterations	200–2000 (integer, step 100)	600
	depth	1–8 (integer, step 1)	4
	learning rate	0.002–0.3 (log)	0.0135
	l2 leaf reg	1–10 (log)	5.77

Table S3: Predictor importance: mean absolute SHAP value of the full model in 500 development-set participants (probability scale, before recalibration).

Rank	Predictor	Mean |SHAP|
1	BMI, kg/m^2	0.0236
2	Weight, kg	0.0189
3	Postmenopausal	0.0177
4	Age, years	0.0175
5	Height, cm	0.0135
6	Waist, cm	0.0117
7	Ever used female hormones	0.0085
8	Non-Hispanic Black	0.0068
9	Income-to-poverty ratio	0.0067
10	Prior hip, wrist or spine fracture	0.0066
11	Mexican American	0.0063
12	Alkaline phosphatase, U/L	0.0055
13	Arthritis	0.0042
14	Cancer	0.0040

Rank	Predictor	Mean |SHAP|
15	25(OH) vitamin D, nmol/L	0.0038
16	Stroke	0.0030
17	Total protein, g/dL	0.0028
18	Creatinine, mg/dL	0.0023
19	Current smoker	0.0023
20	Albumin, g/dL	0.0021
21	Diabetes	0.0020
22	Other Hispanic	0.0019
23	Coronary heart disease	0.0016
24	Education level (1–5)	0.0015
25	Other/multiracial (incl. Asian)	0.0014
26	Cholesterol, mg/dL	0.0014
27	Uric acid, mg/dL	0.0009
28	Former smoker	0.0008
29	Parental hip fracture	0.0008
30	Glucose, mg/dL	0.0007
31	Ever daily glucocorticoid use	0.0006
32	Female sex	0.0005
33	Alcohol $\geq$3 drinks/day	0.0005
34	Calcium, mg/dL	0.0001
35	Phosphorus, mg/dL	0.0001

Table S4: Short form by sex in the temporal validation set: AUROC of the short form, the full model, the three-variable model and OST, and the short form's sensitivity and proportion referred at its development threshold (4.0%).

Sex	n	Cases	Short form	Full model	Age, weight, sex	OST	Sensitivity	Referred
Women	858	72	0.835	0.853	0.834	0.834	0.889	43.6%
Men	1,150	24	0.800	0.820	0.800	0.800	0.583	15.7%

Table S5: Predictors selected on the development set by each feature-selection method (used only for the re-implementation of the published approach).

Method	Selected predictors
Boruta (10)	age, PIR, BMI, weight, height, waist, alkaline phosphatase, creatinine, female sex, postmenopausal
Lasso (34)	all candidates except weight
XGBoost (10)	weight, postmenopausal, age, prior fracture, hormone use, female sex, coronary heart disease, non-Hispanic Black, Mexican American, BMI

Table S6: Re-implementation of the published analysis: 10-fold cross-validation in the development set (mean ± SD across folds). Features were selected once on the full development set, so these estimates are optimistic.

Feature selection	Classifier	AUROC	Accuracy
Boruta	LR	0.871 ± 0.021	0.956
	SVM	0.657 ± 0.050	0.955
	RF	0.849 ± 0.020	0.956
	XGBoost	0.845 ± 0.019	0.954
Lasso	LR	0.879 ± 0.020	0.957
	SVM	0.777 ± 0.042	0.955
	RF	0.848 ± 0.020	0.956
	XGBoost	0.855 ± 0.016	0.956
XGBoost importance	LR	0.872 ± 0.017	0.956
	SVM	0.653 ± 0.086	0.956
	RF	0.854 ± 0.025	0.956
	XGBoost	0.852 ± 0.022	0.955

Table S7: Characteristics of the development (2005–2014) and temporal validation (2017–2018) sets. Mean (SD) or n (%); unweighted tests.

Characteristic	Development (n = 7,649)	Validation (n = 2,008)	P
Osteoporosis (hip)	343 (4.5)	96 (4.8)	
Age, years	64.25 (9.48)	64.07 (8.91)	0.43
Income-to-poverty ratio	2.72 (1.62)	2.69 (1.60)	0.48
Education level (1–5)	3.22 (1.34)	3.48 (1.25)	<0.001
BMI, kg/m^2	28.63 (5.46)	29.14 (5.94)	<0.001
Weight, kg	80.51 (17.70)	81.11 (19.22)	0.21
Height, cm	167.50 (9.97)	166.53 (9.81)	<0.001
Waist, cm	100.95 (13.62)	101.77 (14.39)	0.022
Calcium, mg/dL	9.46 (0.38)	9.32 (0.37)	<0.001
Phosphorus, mg/dL	3.74 (0.56)	3.54 (0.53)	<0.001
Alkaline phosphatase, U/L	72.64 (24.37)	82.57 (25.16)	<0.001
Creatinine, mg/dL	0.99 (0.54)	0.96 (0.41)	0.031
Albumin, g/dL	4.20 (0.30)	4.04 (0.30)	<0.001
Uric acid, mg/dL	5.71 (1.42)	5.66 (1.47)	0.21
Glucose, mg/dL	110.18 (44.03)	109.68 (43.61)	0.66
Total protein, g/dL	7.12 (0.50)	7.12 (0.44)	0.67
Cholesterol, mg/dL	199.23 (43.63)	192.77 (44.14)	<0.001
25(OH) vitamin D, nmol/L	65.06 (25.99)	75.84 (32.93)	<0.001
Female sex	3,293 (43.1)	858 (42.7)	0.81
Diabetes	1,380 (18.1)	468 (23.3)	<0.001
Arthritis	3,027 (39.7)	816 (40.7)	0.41
Stroke	429 (5.6)	128 (6.4)	0.22
Coronary heart disease	570 (7.5)	156 (7.8)	0.67
Cancer	1,181 (15.5)	305 (15.2)	0.80
Former smoker	2,722 (35.6)	621 (30.9)	<0.001
Current smoker	1,338 (17.5)	316 (15.7)	0.067
Prior hip, wrist or spine fracture	917 (12.0)	250 (12.5)	0.59

Characteristic	Development	Validation	P
Parental hip fracture	716 (9.6)	218 (11.0)	0.068
Ever daily glucocorticoid use	390 (5.1)	127 (6.4)	0.032
Postmenopausal	2,889 (38.5)	755 (38.6)	0.95
Ever used female hormones	1,161 (15.5)	208 (10.7)	<0.001
Alcohol ≥3 drinks/day	168 (2.3)	57 (3.0)	0.097
Mexican American	1,118 (14.6)	231 (11.5)	<0.001
Other Hispanic	608 (7.9)	217 (10.8)	<0.001
Non-Hispanic Black	1,549 (20.3)	485 (24.2)	<0.001
Other/multiracial (incl. Asian)	453 (5.9)	357 (17.8)	<0.001

Table S8: Participants aged ≥50 years in the cohort versus those excluded for lack of valid femoral neck DXA. Means or proportions; P from Welch's t-test.

Characteristic	Cohort (n = 9,657)	No DXA (n = 3,185)	Missing in no-DXA group	P
Age, years	64.2	67.6	0.0%	<0.001
Female sex	43.0%	59.1%	0.0%	<0.001
BMI, kg/m^2	28.7	33.0	25.1%	<0.001
Weight, kg	80.6	88.7	24.0%	<0.001
Prior hip, wrist or spine fracture	12.1%	15.6%	0.1%	<0.001
Ever daily glucocorticoid use	5.4%	7.7%	1.4%	<0.001
Non-Hispanic Black	21.1%	24.8%	0.0%	<0.001

Table S9: Percentage of missing values by cycle (final cohort). Predictors not shown had ≤1% missing in every cycle.

Predictor	2005–06	2007–08	2009–10	2013–14	2017–18
Income-to-poverty ratio	4.9	9.8	9.9	8.2	13.4
25(OH) vitamin D, nmol/L	2.6	14.6	4.1	3.2	3.5
Biochemistry panel (range)	3.0–3.2	4.5–4.6	4.7–4.8	4.0–4.3	4.7–4.8
Waist, cm	1.1	1.1	1.1	1.6	1.2
Parental hip fracture	2.6	2.7	2.7	2.4	1.5
Postmenopausal	1.6	1.7	3.0	1.4	2.7
Ever used female hormones	1.8	1.9	3.3	1.6	2.9
Alcohol ≥3 drinks/day	2.9	3.5	5.7	3.9	4.7

Table S10: Survey-weighted performance in the temporal validation set. Weighted: 2017–2018 examination weights, with design-based 95% CIs from a Rao–Wu rescaled bootstrap over primary sampling units within strata (500 replicates). Non-response adjusted: examination weights divided by the modelled probability of having a valid DXA result. Δ, full model minus comparator (non-response-adjusted). Observed prevalence 4.2% (weighted) and 4.2% (non-response adjusted).

Model	Weighted AUROC (95% CI)	Non-response-adjusted AUROC (95% CI)	Δ (95% CI)	Predicted prevalence (NR-adjusted)
Full model	0.889 (0.835–0.937)	0.872 (0.812–0.934)	–	3.9%
Full model without race	0.887 (0.833–0.933)	0.871 (0.802–0.933)	0.001 (−0.004 to 0.009)	3.7%
Boruta + XGBoost pipeline	0.872 (0.809–0.929)	0.860 (0.788–0.927)	0.012 (−0.014 to 0.040)	3.8%
LR: age, weight, sex	0.870 (0.822–0.917)	0.856 (0.793–0.912)	0.016 (−0.009 to 0.043)	3.5%
OST	0.868 (0.816–0.914)	0.849 (0.785–0.913)	0.023 (−0.003 to 0.051)	3.4%
ORAI	0.819 (0.762–0.874)	0.779 (0.688–0.874)	0.093 (0.043–0.147)	3.6%
SCORE	0.860 (0.807–0.904)	0.837 (0.762–0.906)	0.035 (0.004–0.070)	3.8%
LR-spline	0.881 (0.819–0.935)	0.866 (0.793–0.934)	0.006 (−0.007 to 0.019)	4.2%
SVM	0.885 (0.830–0.935)	0.873 (0.815–0.937)	−0.001 (−0.015 to 0.018)	4.0%
Random forest	0.877 (0.810–0.936)	0.870 (0.804–0.934)	0.002 (−0.028 to 0.029)	3.9%
XGBoost	0.883 (0.824–0.938)	0.871 (0.805–0.941)	0.001 (−0.017 to 0.020)	4.0%
LightGBM	0.885 (0.820–0.938)	0.874 (0.802–0.941)	−0.002 (−0.021 to 0.017)	3.9%
CatBoost	0.885 (0.830–0.940)	0.874 (0.812–0.935)	−0.002 (−0.024 to 0.018)	3.9%
Stacking	0.890 (0.836–0.937)	0.877 (0.812–0.940)	−0.005 (−0.018 to 0.006)	3.9%

Table S11: Full model versus OST by sex in the temporal validation set. AUROC difference with paired bootstrap 95% CI. OST cut-off columns show sensitivity / specificity / proportion referred (no fitting). Within each sex, the three-variable model had the same AUROC as OST.

Sex	n	Cases	Full model AUROC	OST AUROC	Δ (95% CI)	OST < 2	OST < -1
Women	858	72	0.853	0.834	0.019 (−0.005 to 0.046)	0.875 / 0.552 / 48.4%	0.694 / 0.847 / 19.8%
Men	1,150	24	0.820	0.800	0.020 (−0.031 to 0.068)	0.833 / 0.689 / 32.2%	0.500 / 0.896 / 11.2%

Table S12: Full model at alternative operating points in the temporal validation set (96 cases). The first two thresholds were chosen on the development set; the others are fixed predicted risks.

Operating point	Threshold	Referred	Detected	Sens.	Spec.	PPV	NPV
Youden (development)	4.8%	23.8%	77/96	0.802	0.791	0.161	0.988
Sensitivity 0.90 in development	2.3%	39.3%	86/96	0.896	0.632	0.109	0.992
Fixed 3%	3.0%	32.8%	83/96	0.865	0.699	0.126	0.990
Fixed 5%	5.0%	22.8%	76/96	0.792	0.801	0.166	0.987
Fixed 10%	10.0%	12.0%	63/96	0.656	0.907	0.261	0.981

Table S13: Net benefit in the temporal validation set at selected threshold probabilities (95% bootstrap CIs), with true and false positives per 100 people screened. OST recalibrated to probabilities.

Model	Threshold	Net benefit (95% CI)	TP per 100	FP per 100
Full model	0.05	0.0279 (0.0194–0.0371)	3.78	18.97
	0.10	0.0215 (0.0148–0.0299)	3.14	8.86
	0.15	0.0122 (0.0059–0.0191)	2.19	5.48
	0.20	0.0107 (0.0046–0.0171)	1.84	3.09
Boruta + XGBoost	0.05	0.0259 (0.0176–0.0341)	3.64	19.92
	0.10	0.0190 (0.0122–0.0268)	2.89	8.91
	0.15	0.0146 (0.0071–0.0220)	2.44	5.53
	0.20	0.0078 (0.0022–0.0141)	1.69	3.64
LR: age, weight, sex	0.05	0.0265 (0.0181–0.0354)	3.69	19.62
	0.10	0.0181 (0.0106–0.0256)	2.84	9.26
	0.15	0.0114 (0.0056–0.0178)	1.89	4.28
	0.20	0.0078 (0.0025–0.0139)	1.44	2.64
OST (recalibrated)	0.05	0.0268 (0.0181–0.0356)	3.74	20.12
	0.10	0.0183 (0.0108–0.0261)	2.84	9.11
	0.15	0.0077 (0.0021–0.0140)	1.69	5.23
	0.20	0.0056 (0.0007–0.0112)	1.29	2.94
ORAI (recalibrated)	0.05	0.0202 (0.0128–0.0280)	3.09	20.27
	0.10	0.0122 (0.0064–0.0188)	1.99	6.97
	0.15	0.0048 (0.0011–0.0091)	0.80	1.79
	0.20	0.0035 (-0.0002–0.0080)	0.80	1.79
SCORE (recalibrated)	0.05	0.0257 (0.0177–0.0345)	3.64	20.22
	0.10	0.0160 (0.0091–0.0231)	2.59	8.91
	0.15	0.0085 (0.0031–0.0144)	1.64	4.48
	0.20	0.0054 (0.0001–0.0112)	1.29	3.04

Table S14: Sensitivity analyses: temporal-validation AUROC and calibration slope.

Analysis / model	Dev. prev.	Val. prev.	AUROC	Slope
Hip or complete-scan lumbar spine (L1–L4)				
Full model	7.2%	6.7%	0.846	1.06
OST			0.823	1.11
LR: age, weight, sex			0.836	1.12
Hip or lumbar spine from ≥2 valid vertebrae				
Full model	8.4%	8.2%	0.840	1.05
OST			0.801	1.04
LR: age, weight, sex			0.824	1.10
Excluding non-Hispanic Asian participants				
Full model	4.4%	4.4%	0.861	0.92
Development on 2013–2014 only (same scanner type)				
Full model	4.8%	4.8%	0.854	1.13
Development set without 2005–2006				
Full model	4.5%	4.8%	0.869	1.00
Without laboratory predictors				
Full model	4.5%	4.8%	0.863	0.98
Without alcohol predictor				
Full model	4.5%	4.8%	0.869	0.99
Including previously diagnosed or treated participants				
Full model	6.4%	7.1%	0.859	0.97
OST			0.824	0.89
LR: age, weight, sex			0.839	0.95

Table S15: Coefficients of the full model (penalised logistic regression on 35 predictors, recalibrated, original units). Units as in Supplementary Table S7; binary predictors 1 = yes. Coefficients are penalised and correlated predictors are not individually interpretable.

Term	Coefficient	Odds ratio per unit
Intercept	6.95527	
Age, years	0.05908	1.061
Income-to-poverty ratio	−0.14411	0.866
Education level (1–5)	0.03807	1.039
BMI, kg/m^2	−0.15810	0.854
Weight, kg	−0.03937	0.961
Height, cm	−0.04479	0.956
Waist, cm	0.02589	1.026
Calcium, mg/dL	−0.01321	0.987
Phosphorus, mg/dL	−0.00500	0.995
Alkaline phosphatase, U/L	0.00771	1.008
Creatinine, mg/dL	0.29171	1.339
Albumin, g/dL	−0.27440	0.760
Uric acid, mg/dL	−0.02154	0.979
Glucose, mg/dL	−0.00101	0.999
Total protein, g/dL	−0.20078	0.818
Cholesterol, mg/dL	0.00119	1.001

Term	Coefficient	Odds ratio per unit
25(OH) vitamin D, nmol/L	−0.00539	0.995
Female sex	−0.02615	0.974
Diabetes	0.18337	1.201
Arthritis	−0.24444	0.783
Stroke	0.53830	1.713
Coronary heart disease	0.31449	1.370
Cancer	−0.38293	0.682
Former smoker	−0.05355	0.948
Current smoker	0.20477	1.227
Prior hip, wrist or spine fracture	0.93243	2.541
Parental hip fracture	−0.14696	0.863
Ever daily glucocorticoid use	0.14705	1.158
Postmenopausal	0.99824	2.714
Ever used female hormones	−0.79702	0.451
Alcohol ≥3 drinks/day	−0.33579	0.715
Mexican American	−0.77710	0.460
Other Hispanic	−0.36512	0.694
Non-Hispanic Black	−0.80116	0.449
Other/multiracial (incl. Asian)	−0.26758	0.765

Table S16: Temporal-validation results in every version of this analysis. Versions differed in outcome definition, cohort and comparators, as listed; all used the same 2005–2014 / 2017–2018 split. The present analysis is version 6.

Version	Outcome	Cohort and predictors	Prevalence	Best model AUROC (95% CI)	OST	LR: age, weight, sex
1	Femoral neck or L1–L4 (complete scans)	All with DXA; 29 predictors	9.2%	LR 0.843 (0.813–0.871)	–	–
2	Femoral neck or spine (≥2 vertebrae)	All with DXA; 35 predictors	10.8%	LR 0.839 (0.813–0.864)	0.796	0.822
3	Hip (femoral neck or total hip)	All with DXA; 35 predictors	6.6%	LR-spline 0.869 (0.839–0.895)	0.824	0.839
4	Hip	Previously diagnosed excluded; 35 predictors	4.7%	LR 0.870 (0.835–0.905)	0.839	0.850
5	Hip	Diagnosed or treated excluded; all families tuned	4.5%	LR 0.869 (0.831–0.903)	0.841	0.851
6	Hip	As 5; 40 trials for every family, wider search spaces; ORAI, SCORE; IECV	4.5%	LR 0.869 (0.831–0.904)	0.841	0.851

Table S17: TRIPOD+AI checklist [17]: where each item is reported.

Item	Topic	Where reported
1	Title	Title page
2	Abstract	Abstract
3a	Healthcare context and rationale	Introduction, paragraphs 1–3
3b	Intended purpose and users	Introduction; Discussion, Implementation considerations
3c	Known health inequalities	Introduction (item 5); Methods, Subgroup and fairness analysis
4	Objectives	Introduction, final paragraphs
5a	Data sources	Methods, Data source
5b	Dates of data collection	Methods, Data source; Development and temporal validation sets

Item	Topic	Where reported
6a	Setting	Methods, Data source
6b	Eligibility criteria	Methods, Study population; Figure 1
6c	Treatments received	Methods, Study population (diagnosed or treated excluded)
7	Data preparation and quality checks	Methods, Data preparation; Table S9
8a	Outcome definition and assessment	Methods, Outcome definition
8b	Outcome assessors	Methods, Outcome definition (NHANES DXA technologists)
8c	Blinding of outcome assessment	Methods, Outcome definition (DXA independent of predictors)
9a	Choice of predictors	Methods, Candidate predictors
9b	Predictor definitions and measurement	Methods, Candidate predictors; Data preparation
9c	Predictor assessors and blinding	Standard NHANES interview, examination and laboratory protocols
10	Sample size	Methods, Sample size
11	Missing data	Methods, Missing data; Table S9
12a	Use of data in development and evaluation	Methods, Development and temporal validation sets
12b	Handling of predictors	Methods, Missing data; Models (splines, standardisation)
12c	Model type, building and tuning	Methods, Models, class-imbalance strategies and tuning; Table 2; Supplementary Table S2
12d	Heterogeneity across clusters	Methods, Internal–external validation; Sensitivity analyses (cycles, scanner); Survey weighting and non-response
12e	Performance measures	Methods, Performance evaluation
12f	Model updating or recalibration	Methods, Recalibration and thresholds
12g	How predictions are calculated	Methods, Short-form model; Tables 9 and S15; code
13	Class imbalance	Methods, Models, class-imbalance strategies and tuning
14	Fairness	Methods, Subgroup and fairness analysis
15	Model output and thresholds	Methods, Recalibration and thresholds; Supplementary Table S12
16	Differences between development and evaluation data	Results, Participant characteristics; Table S7
17	Ethics	Declarations
18a–b	Funding; conflicts of interest	Declarations
18c–d	Protocol; registration	Declarations (none)
18e–f	Data and code availability	Declarations
19	Patient and public involvement	Declarations (none)
20a	Participant flow	Figure 1
20b	Participant characteristics	Table 1
20c	Comparison of development and evaluation data	Table S7

Item	Topic	Where reported
21	Numbers of participants and outcome events	Figure 1; Methods, Sample size; Table 7
22	Full model specification	Tables 9 and S15; code repository
23a	Performance with confidence intervals, including subgroups	Tables 3, 7 and 8; Figures 2–4
23b	Heterogeneity across clusters	Table 4; Supplementary Tables S14, S10 and S11
24	Model updating results	Results, Calibration and clinical utility
25	Interpretation	Discussion
26	Limitations	Discussion, Strengths and limitations; Supplementary Table S16
27a	Handling of poor-quality or missing input data	Methods, Missing data (imputation model in code)
27b	User interaction and required expertise	Discussion, Implementation considerations
27c	Future research and applicability	Discussion, Implementation considerations; Limitations

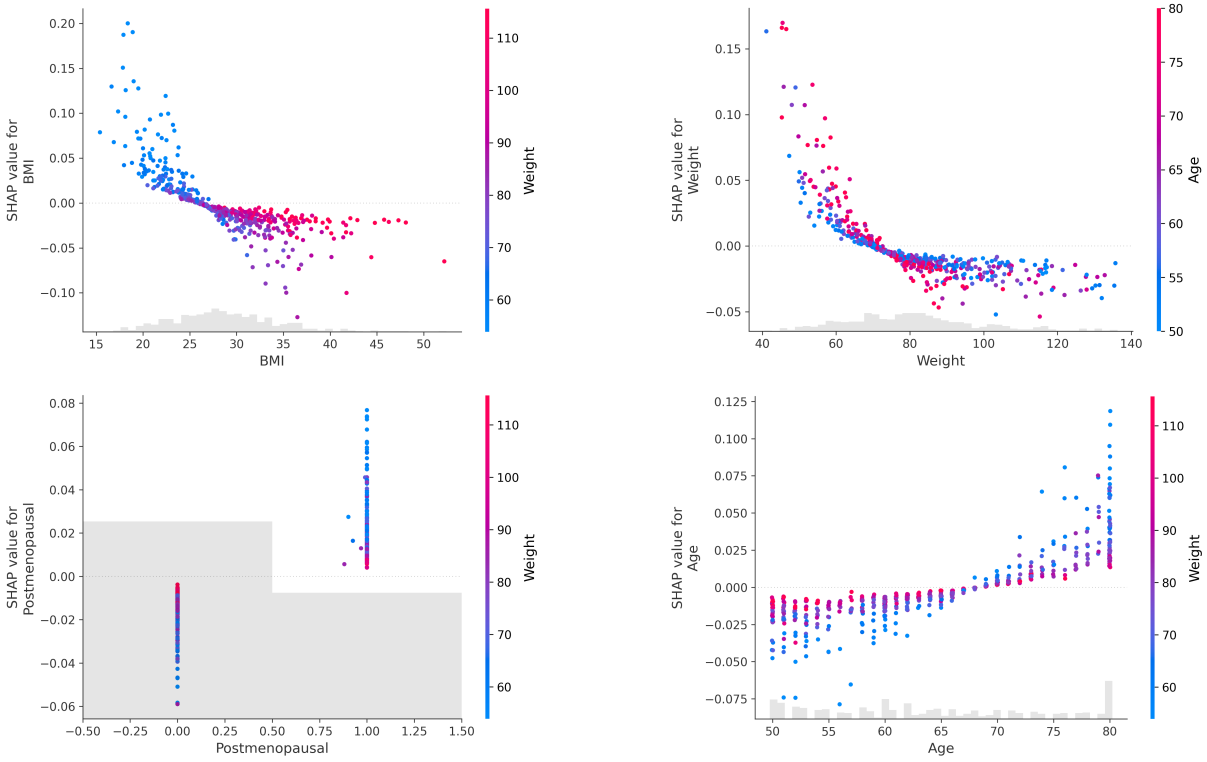


Figure S2: SHAP dependence plots for the four most important predictors (BMI, weight, postmenopausal status, age), full model, 500 development-set participants.

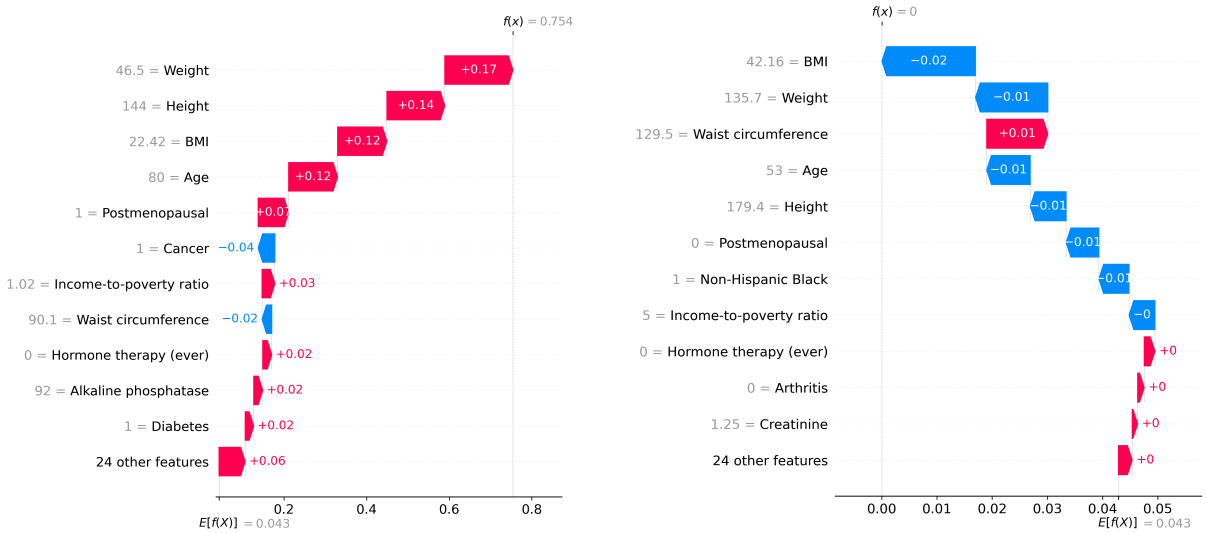


Figure S3: SHAP waterfall plots for the highest-risk (left) and lowest-risk (right) individuals in the explained development sample.